Access is only the first mile

Mobile money and electricity uptake across developing countries, climate-resilient primary care in Uganda, and a behavioral-development classic from Ethiopia · two peer-reviewed economics articles plus one rigorous World Bank working paper, all represented by lawful open editions. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

Across electricity, health, and agriculture, access to an asset or service is only the first mile. Mobile money helps households turn grid proximity into an actual electricity connection. Community health workers turn primary-care availability into protection against rainfall-driven infant mortality. In Ethiopian villages, social preferences can make even useful farm innovations costly to adopt. **Literature trend:** applied development economics increasingly locates the binding constraint between formal provision and practical use—in payment systems, local delivery capacity, and the social incentives surrounding adoption. Building the thing matters; making it usable, resilient, and socially viable often determines the effect.

01 CURRENT / INFRASTRUCTURE, FINANCE, AND TECHNOLOGY ADOPTION · STARTS ON PAGE 2 · World Bank Policy Research Working Paper 11492 · 2026 · CC BY 3.0 IGO

Financial Inclusion and Electricity Uptake: Global Evidence and District-Level Mechanisms

Why do households remain disconnected when the electricity grid is already nearby? Lang and Ly exploit staggered mobile-money introductions across 180 non-OECD countries using a heterogeneity-robust difference-in-differences estimator, then add district-level evidence from 33 sub-Saharan African countries using mobile-network coverage and entropy balancing. Mobile money is associated with higher electrification, especially in rural areas, and roughly 24% greater power uptake in comparable covered districts; remittances and lower payment frictions fit the mechanism, while new generation capacity does not. HonestDiD sensitivity tests and alternative weighting methods strengthen the result, but adoption timing and district access remain nonrandom. The paper makes infrastructure's last mile concrete: financial rails can determine whether physical rails are actually used.

02 CURRENT / CLIMATE ADAPTATION AND GLOBAL HEALTH · STARTS ON PAGE 17 · Journal of Development Economics 176 · 2025 · peer reviewed · CC BY 4.0

Weather Shocks, Infant Mortality, and Adaptation: Experimental Evidence from Uganda

Can basic primary care make poor households more resilient to climate shocks? The authors combine randomized placement of a community-health-worker program across 214 Ugandan villages with independently measured growing-season rainfall. Infant mortality rises after rainfall deficits in control villages but not in treated villages; at the moderate-drought threshold, access to community health workers lowers infant death risk by 46%. The result survives alternative rainfall definitions, mortality windows, spatial standard errors, and checks on migration and baseline balance. Treatment spillovers, relatively few deaths, and an inability to isolate one precise mechanism temper the estimate. Still, the design offers unusually direct evidence that health-system capacity is climate adaptation infrastructure.

03 CLASSIC / BEHAVIORAL DEVELOPMENT AND AGRICULTURAL INNOVATION · STARTS ON PAGE 32 · World Development 67 · 2015 · peer reviewed classic · CC BY 3.0

Social Preferences and Agricultural Innovation: An Experimental Case Study from Ethiopia

Can envy or spite slow the diffusion of useful farm technologies? Kebede and Zizzo run money-burning games with Ethiopian farmers, randomizing initial endowments and lottery outcomes so participants can pay to reduce anonymous peers' earnings. Behavior fits an inequality-aversion model based on absolute payoff differences better than reciprocity or relative-to-average alternatives. Villages with more money burning also show lower adoption of fertilizer, improved seed, and rainwater harvesting, including in an independent earlier survey. The game cleanly identifies motives inside the experiment, but the bridge to real-world innovation is correlational and drawn from a limited set of villages. The paper is a memorable warning that technology adoption can carry social as well as private costs.

Editorial note: references and appendices were omitted where possible; no included article contributes more than about 15 text pages, excluding table-heavy pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

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Policy Research Working Paper 11492

Financial Inclusion and Electricity Uptake

Global Evidence and District-Level Mechanisms

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WORLD BANK GROUP
Development Economics
Development Research Group
October 2026



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Policy Research Working Paper 11492

Abstract

Even as governments push to build infrastructure to achieve universal access to electricity, demand-side barriers constrain uptake where infrastructure already exists. This paper assess the impact of the quasi-experimental introduction of mobile money on electricity adoption by households. Leveraging a cross-country panel of 180 non-Organisation for Economic Co-operation and Development (OECD) countries, it first employs a heterogeneous difference-in-differences estimator to document a significant and positive association between mobile money introduction and electrification, particularly in rural areas. The global analysis is complemented with a granular district-level analysis of 33 sub-Saharan African countries. The district-level analysis leverages differential sub-national mobile network coverage in addition to country-level mobile-money adoption. The results indicate that mobile money access improves district-level power uptake by around 24% relative to the similar districts without mobile money access. Supply-side infrastructure expansion is ruled out as the primary driver. Instead, evidence suggests demand-side channels, specifically reduced financial frictions and increased remittances, drive this relationship. The results highlight the enabling role of mobile network coverage and, conversely, the detrimental impact of mobile money taxes on electrification efforts.

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Produced by the Research Support Team

Financial Inclusion and Electricity Uptake: Global Evidence and District-Level Mechanisms

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Reproducibility package: https://reproducibility.worldbank.org/catalog/573

Authorized for distribution by Florence Kondylis, Research Manager, Development Research Group, Development Economics, World Bank Group

JEL Classification: O16, O33, Q40

Keywords: Mobile money, Financial inclusion, Electrification, Demand-side factors

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1 Introduction

Electricity access rates remain low throughout many developing regions, particularly in sub-Saharan Africa, relative to other regions of the world. While supply-side constraints explain low access in remote areas, electrification rates often remain low in areas already covered by the grid. Low under-grid access rates highlight the importance of demand-side factors: households may face frictions such as credit constraints, poor information, or high transaction costs for paying electricity bills. Given limited government budgets, understanding to what extent other policies can generate complementary gains in electrification is a critical question.

Recent decades have seen major gains in financial inclusion, notably through the introduction of mobile money solutions. Evidence suggests that mobile money can help households smooth consumption, lower transaction costs, increase remittances, and promote savings. Given that mobile money lowers financial frictions, is it an effective tool for promoting electrification?

This paper combines two distinct empirical approaches to answer this question. First, we leverage cross-country variation in the timing of mobile money adoption across 180 non-OECD countries from 1990–2021. Using the heterogeneous difference-in-differences estimator proposed by Callaway and Sant'Anna (2021), we find that mobile money is associated with significant increases in urban electrification rates within one year and significant increases in rural electrification rates after five years.

Second, to address identification challenges related to the difficulty of disentangling demand-side and supply-side barriers, we zoom in on sub-Saharan Africa using a dataset of 2,825 districts combined with demographic and health surveys (DHS). These data allow us to exploit the distinction between households covered by the grid and those not covered, as well as within-country variation in mobile network coverage, to better understand the relationship between mobile money and electrification. Using entropy balancing to construct comparable control groups and an instrumental variables strategy, we find that districts with mobile money access have 24% higher electrification rates than comparable districts without mobile money access.

Our work broadly contributes evidence on drivers of demand for electrification in low-income settings. Although energy access programs have successfully expanded grid coverage, many households remain unelectrified (Barry and Creti, 2020; Sievert and Steinbuks, 2020; Ankel-Peters et al., 2025). Much of the existing evidence exploring this gap focuses on specific demand-side constraints. For instance, multiple studies consider consumer responses to changes in the price of electrification (Chaplin et al., 2017; Lee et al., 2020; Grimm et al., 2020; Burgess et al., 2025; Lang, 2026). Others study the role of information frictions (Bernard and Torero, 2015; Urpelainen and Yoon, 2015; Alem and Dugouan, 2022; Mahadevan et al., 2023; Blankenship et al., 2020) or the anticipated quality of electricity (Kennedy et al., 2019; Bajo-Buenestado, 2021; Coville et al., 2025). However, addressing these factors individually often yields incomplete uptake because low-income households face interlocking financial barriers. As Dugoua et al. (2017), Blimpo and Cosgrove-Davies (2019), and Blimpo et al. (2020) point out, households must simultaneously navigate high upfront connection costs and wiring costs while managing regular bill payments with uncertain income flows.

Mobile money directly targets financial frictions. Although it does not directly reduce the costs associated with electrification, it can help households navigate financial frictions like credit constraints and

uncertain income flows, and even lead to increased incomes for households who receive regular remittance payments. Our work complements two studies that use data from the World Bank's Multi-tier Framework to quantify the relationship between mobile money use and electrification (Angelou et al., 2026). First, Mahn et al. (2025) show that mobile banking significantly increases willingness to pay for a hypothetical grid connection or a hypothetical solar home system when payments need to be made in fewer than six months, but not under longer payment horizons. Our work complements their results by providing evidence on real-world connection rates under different mobile money regimes. Second, Mbate and Fall (2025) document that individuals with mobile money accounts in Kenya are significantly more likely to connect to the grid than those without.[1] Our results expand this analysis throughout Africa, which is critical given the unusually high speed of adoption of mobile money in Kenya. Most importantly, by exploiting the *introduction* of mobile money networks rather than individual adoption decisions, our empirical strategy avoids the selection issues inherent in comparing mobile money users to non-users. By performing our analysis at scale across the continent and focusing on mobile money rollouts and network coverage, we provide robust evidence that mobile money can help drive electrification.

Our results also contribute to a growing base of evidence on the impacts of mobile money. It is well-established that mobile money leads to increases in remittances across long distances, allowing households to better manage shocks and smooth consumption (Jack and Suri, 2014; Munyegera and Matsumoto, 2016; Apiors and Suzuki, 2018; Apeti, 2023; Riley, 2018). In addition, alternative evidence suggests that mobile money may contribute to increased agricultural productivity, movement out of agriculture, increased urban migration, reductions in poverty, and increased tax revenue (Jack et al., 2013; Suri and Jack, 2016; Sekabira and Qaim, 2017; Aron, 2018; Batista and Vicente, 2020; Apeti and Edoh, 2023). Our results provide novel evidence suggesting that financial inclusion and lower transaction costs also contribute to increased electricity uptake. Notably, we can show that the converse is also true: increasing transaction costs through taxes on mobile money reverses gains in electrification. As such, our results provide evidence about which types of electrification policy goals may be best supported by concurrent advances in financial inclusion and which conditions yield the greatest complementary effects.

2 Conceptual framework or potential mechanisms

Although it is clear why electrification may lead to increased demand for mobile money (and mobile phones in general), the causal link from mobile money access to households' electrification decisions is less obvious. In this section, we outline potential demand-side causal pathways linking mobile money to electrification. We are limited in our ability to distinguish between these pathways empirically, but we provide suggestive evidence when discussing our results.

Figure 1 shows three causal pathways from mobile money access to power uptake at the household level. Each pathway is not mutually exclusive: all three may play a role.

The first causal pathway focuses on income. In theory, households with mobile money are able to rely on the easy and affordable transfer of money from family and friends to help smooth consumption across negative shocks. Smoother consumption and this type of informal insurance may allow households to invest in higher risk, higher return activities. Permanent flows of remittances from family members or transfers

[1] In the same vein, Barry et al. (2025) find that mobile money services have a positive effect on solar panel adoption in Tanzania.

treatment effects across groups or when treatment effects are dynamic (see Callaway and Sant'Anna, 2021; De Chaisemartin and d'Haultfoeuille, 2020; Sun and Abraham, 2021). As we have multiple periods and multiple treatment timings (2002–2010, 2011, 2012, 2013–2015, 2016–2021), we cannot ensure that the treatment effects are homogeneous across both groups and time. We therefore move forward with the estimator proposed by Callaway and Sant'Anna (2021).

Callaway and Sant'Anna (2021) offer a staggered DiD framework to identify the causal effect of treatments with variation in treatment timing. The estimator has the ability to conduct robust inference on treatment effect heterogeneity resulting from the staggered exposure to treatment. It also allows for simultaneous calculation of confidence intervals that adjust for multiple hypothesis testing. Let G_g be a binary variable denoting the cohort of observations first treated at time g; and G_∞ is a binary variable representing observations not treated by the last time period in our data. If pre-treatment covariates do not affect the identification, then the average treatment effect on the treated (ATT) for cohort g at time t is calculated as:

$$ATT(g,t) = \mathbb{E}\left[Y_t - Y_{g-1} \mid G_g = 1\right] - \mathbb{E}\left[Y_t - Y_{g-1} \mid G_\infty = 1\right]. \tag{2}$$

For every g, the first term measures the expected difference between outcome Y_t and Y_{g-1} (the period immediately before treatment), conditional on being in cohort g. Focusing on two time periods (t and $g-1$) thus captures variation in the treatment effect resulting from the differences in first exposure and the length of exposure. The second term is the counterfactual, using the never treated cohort as the comparison group.[4]

Factors such as population density (see for instance Van Ruijven et al., 2012; Trotter, 2016; Bertheau et al., 2017; Khavari et al., 2021a; or Khavari et al., 2021b), households' income (e.g. Louw et al., 2008; Jimenez, 2017; Aklin et al., 2018; Shibano and Mogi, 2020) or even inequality (Sarkodie and Adams, 2020a; Sarkodie and Adams, 2020b; Aklin et al., 2021; Murshed, 2023; Rustagi et al., 2024) might affect electrification. Higher population density may result in easier access to electricity as it becomes more feasible for utilities to reach a larger number of consumers within a smaller geographic area. In contrast, remote areas with low population density might be harder to reach, making efforts to expand electrification more challenging and costly. Similarly, households with higher incomes are more likely to afford the upfront costs associated with connecting to the grid and are better positioned to afford regular bills payment for electricity consumption. Finally, inequality often manifests as disparities between urban and rural areas. Urban areas, where higher-income populations typically reside, tend to have better access to multiple types of infrastructure compared to rural and remote areas, leading to urban-rural gaps in electrification rates.

In our context, we have opted to retain the same covariates for both electrification (outcome model) and mobile money (treatment model), given that both are forms of infrastructure. Although we lack existing literature on the predictors of mobile money adoption, we can draw parallels with how population density might influence investment in electricity infrastructure. It stands to reason that population density could also impact investment in mobile networks, with operators likely to prioritize countries with higher population densities. Additionally, household income levels are pertinent factors for operators assessing market potential and demand, whether for electricity or mobile services. Moreover, inequality might in-

[4] In our baseline specification, the comparison group consists of never-treated units, defined as $G_\infty = 1$. As a robustness check, we alternatively use the not-yet-treated cohorts as the comparison group, defined as units with $G_h = 1$ and $h > t$, i.e. $\mathbb{1}(G_i = h)$ with $h > t$. These correspond to countries that have not yet adopted mobile money at time t but may adopt in a future period. The resulting ATT(g,t) estimates and their aggregation remain quantitatively and qualitatively similar, confirming that our results are not driven by the choice of comparison group.

centivize governments to pursue inclusive policies, thereby promoting financial inclusion through mobile money initiatives.

Including these factors as time-varying covariates in our estimates allows us to obtain conditional parallel trends. With pre-treatment covariates, the $\text{ATT}(g,t)$ is calculated using the AIPW (Augmented Inverse Probability Weighting) estimator as:

$$ATT(g,t) = \mathbb{E}\left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_g(X)G_\infty}{1-p_g(X)}}{\mathbb{E}\left[\frac{p_g(X)G_\infty}{1-p_g(X)}\right]}\right)(Y_t - Y_{g-1} - m_{g,t}(X))\right] \tag{3}$$

where $m_{g,t}(X) = \mathbb{E}[Y_t - Y_{g-1} \mid C = 1]$. In this case, the doubly robust estimator (see Sant'Anna and Zhao, 2020) uses two cohorts G_g and G_∞ with stabilized weights in the denominator to ensure the weights integrate to 1. The term $p_g(X)$ is a generalized propensity score modeled using a logistic function that weights the observation by the probability that the observation is placed in the treated group. The last set of terms is a DiD with recentered covariates using the outcome regression on never treated groups, $m_{g,t}(X)$, also called the augmented term.

Our coefficient of interest γ^{CS} estimated using the Callaway and Sant'Anna (2021) DiD is an aggregation of the weighted mean of all ATTs described above.

$$\gamma^{\text{CS}} = \sum_{g\in G}\sum_{t=2}^{T} \underbrace{w(g,t)}_{\text{weight}} \text{ATT}(g,t) = \frac{1}{\kappa}\sum_{g\in G}\sum_{t=2}^{T} \mathbb{1}(t \geq g)\Pr(G = g \mid G \leq T)\,\text{ATT}(g,t) \tag{4}$$

where $\kappa = \sum_{g\in G}\sum_{t=2}^{T} \mathbb{1}(t \geq g)\Pr(G = g \mid G \leq T)$ ensures that the weights on $\text{ATT}(g,t)$ sum up to 1. The coefficient of interest γ^{CS} is a weighted average of each $\text{ATT}(g,t)$ putting more weight on $\text{ATT}(g,t)$ with larger group sizes. Unlike the estimation of γ^{TWFE} in the TWFE specification, this combination of $\text{ATT}(g,t)$ to estimate γ^{CS} rules out issues arising from negative weights.

To compute the duration of exposure ATTs, let $e = t - g$ be the length of exposure to the treatment.

$$\gamma^{\text{CS}}(e) = \sum_{g\in G} \underbrace{\mathbb{1}(g + e \leq T)\Pr(G = g \mid g + e \leq T)}_{\text{proportions used to estimate ATT(g, g+e)}} \text{ATT}(g, g+e) \tag{5}$$

Put otherwise, our causal identification relies on the following key assumptions. First, we should not have any treated country in the first period (in 1990 in our case). This assumption is valid as the first adoption occured in 2002. Second, we require common support for propensity scores. Third, the treatment should be irreversible, which holds as we observe no instance of a country dis-adopting mobile money (once treated, always treated). Fourth, we expect no treatment anticipation or limited treatment anticipation in pre-treatment periods. We show evidence of no anticipation graphically. It is also worth noting that many of the mechanisms we consider such as remittances, bill payment, or a higher value of having electricity are not conducive to anticipation effects. Finally, we require parallel trends conditional on the covariates that we include. We test the validity and robustness of the parallel pre-treatment trends assumption using event study figures that show no evidence of pre-trends. However, parallel pre-trends do not necessarily imply parallel counterfactual post-treatment trends. For instance, other policies might change at the same time as mobile money is introduced. We undertake a series of empirical tests to ensure that our results are not driven by concurrent policy changes.

3.3 Results

3.3.1 TWFE estimates versus Heterogeneous DiD

Results from Equation 1 show that there is a significant and positive link between the adoption of mobile money solutions and electrification rates after controlling for population density, the percentage of GDP going to household consumption, wealth inequality, country FEs and time FEs (see Table 1). Our results indicate a positive and significant relationship between mobile money adoption and electrification rates (total, rural, and urban) and a negative and significant relationship between mobile money adoption and the urban-rural electrification gap from 1990–2021. The estimated coefficients suggest that, on average, the adoption of mobile money leads the total electrification rate to increase by 25%, the rural electrification rate to increase by 39%, the urban electrification rate to increase by 11%, and the urban-rural electrification gap to decrease by 28%. Results from the TWFE approach suggest that mobile money adoption may act as a catalyst for improving electrification rates, particularly in rural areas. This aligns with the traditional focus of mobile money services, which often target under-served populations in rural and remote areas.

Table 1: TWFE estimates: Mobile money adoption and electrification

	(1) Total electrification	(2) Rural electrification	(3) Urban electrification	(4) Electrification Gap
Mobile money adoption	0.250***	0.386***	0.114***	-0.276***
	[0.042]	[0.071]	[0.023]	[0.061]
Observations	3007	2995	3003	2995
Adjusted R^2	0.363	0.340	0.266	0.277
ρ	0.911	0.909	0.873	0.894
Country FE	√	√	√	√
Year FE	√	√	√	√

Dependent variables are the log of the total electrification rate, rural electrification rate, urban electrification rate, and rural-urban electrification gap (columns (1) - (4), respectively). Effects are estimated using the specification Equation 1. Robust standard errors in brackets. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Considering the limitations of the TWFE estimator and the large coefficients obtained, we next move to our preferred specification: the heterogeneous Difference-in-Differences (DiD) estimator (Callaway and Sant'Anna, 2021). The heterogeneous DiD estimator allows for more flexibility in capturing the varying effects of mobile money adoption across different groups and eliminates concerns about negative weighting.

Table 2 shows that the results using the estimator from Callaway and Sant'Anna (2021) are qualitatively similar to those from the TWFE estimator; however, the magnitude of the effects are smaller, indicating an upward bias from the standard TWFE estimator. As described in Roth et al. (2023), the standard TWFE estimator produces biased estimates primarily when different units are treated at different times and there is heterogeneity in treatment effects over time. This bias relates to the insight in Goodman-Bacon (2021) that negative weighting of treatment effects arises when already-treated units serve as controls.

Despite the lower magnitudes of the estimated effects, they remain economically meaningful and the fundamental conclusion remains intact: mobile money adoption increases electrification rates and reduces the urban–rural electrification gap. Taking an average across all post-adoption periods indicates an increase of 12% in the electrification rate (6% in the urban electrification rate, and 18% in the rural electrification

rate). The urban-rural electrification gap closes by 12% (see Table 2).

Table 2: Heterogeneous DiD: Mobile money adoption and electrification

	(1) Total electrification	(2) Rural electrification	(3) Urban electrification	(4) Electrification Gap
Mobile money adoption	0.121***	0.176**	0.060**	-0.116*
	[0.034]	[0.055]	[0.021]	[0.047]
Observations	1908	1901	1901	1901

Dependent variables are the log of the total electrification rate, rural electrification rate, urban electrification rate, and rural-urban electrification gap (columns (1) - (4), respectively). Effects are aggregated across all mobile money adopting countries and all time periods, using the specification Equation 4. Standard errors in brackets clustered at the country level. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Although averaging across all post-adoption periods speaks to the aggregate link between mobile money adoption and electrification, Figure 4 illustrates the dynamics of the relationship over time. Importantly, the figure shows no evidence of pre-treatment trends in electrification rates or the urban-rural electrification gap between countries that eventually adopt mobile money and those that do not: the pre-treatment differences in the outcomes between the groups are close to zero and statistically insignificant in the six-year pre-period we observe. When observing the dynamic treatment effects in the post-treatment period, we observe significant effects on the overall electrification rate from the second year after the introduction of mobile money. However, this masks heterogeneity in the timing of effects between urban and rural areas. In urban areas, the effect begins to appear in the first year after adoption, while we observe no significant increases in rural electrification rates until five years after adoption. These differences in timing carry through to reductions in the urban-rural electrification gap, where we observe no significant effects until eight years after adoption. Although these lags are likely due to somewhat unsystematic deployment of mobile money solutions in more remote, rural areas, they do make it more difficult to cleanly attribute the increases in electrification to mobile money alone.

3.3.2 Robustness

Our baseline results remain robust to four additional checks (see Figure A1), which address potential concerns related to the choice of control group, estimator, and sample period. First, we extend the control group to include not-yet-treated countries, rather than relying solely on never-treated countries. This ensures that the results are not sensitive to the composition of the comparison group. Second, we implement the Inverse Probability Weighting (IPW) estimator. The IPW estimates are very similar to our baseline results. Third, we restrict the period of analysis to 1995–2021 to exclude early years with more limited data availability and potentially different electrification dynamics. Fourth, we further restrict the sample to 2000–2021, focusing on the period immediately preceding the first introduction of mobile money. Across all these specifications, the magnitude and direction of the estimated effects remain stable.

Moreover, mobile money adoption varies across countries in our sample, with some adopting the service as early as 2002 while others have adopted it only recently. Understanding these differences is crucial for accurate analysis. To ensure robust estimation, we apply additional criteria and conduct further robustness checks. Specifically, we restrict our analysis to countries with at least 8 years of post-adoption data. This

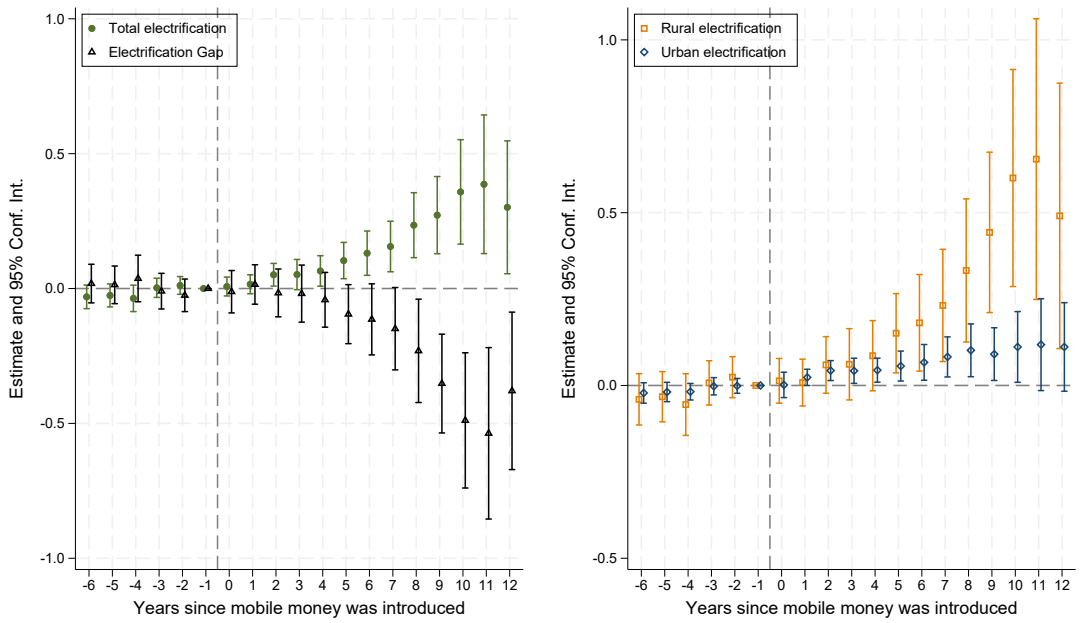


Figure 4: The effects of mobile money introduction on electrification rates (total, rural, urban) and electrification gap, Baseline estimates.

Notes: Event study results based on Equation 5. We show point estimates along with 95% confidence intervals derived from standard errors clustered at the country level. The dependent variables include annual log-transformed electrification rates (total, rural, urban), as well as the electrification gap. The "years from mobile money introduction" variable denotes the length of exposure to mobile money services, measured in years from the date of the first adoption of these services. This variable helps assess the impact of mobile money adoption on electrification rates over time, allowing for a comprehensive understanding of the treatment effect dynamics. To help in understanding the magnitude and significance of the effects, Table A19 presents the detailed results of the event study. Table 2 summarizes the aggregate effects based on Equation 4 to provide a concise overview of the overall impact of mobile money adoption on electrification over the full post-adoption period.

ensures that all control groups are present in each period, allowing for accurate comparisons. The aggregate effects are close to those found in the baseline (see Table A6). Additionally, in our event study analysis (see Figure A2), we focus on an 8-year post-adoption period instead of the 16-year window used in the baseline specification. This adjustment enhances the reliability of our results by providing a more balanced and comparable time window across cohorts and a more precise examination of the effects of mobile money adoption on electrification.

Furthermore, in Figure A3 (full sample) and Figure A4 (sample restricted to countries with at least 8 years under treatment), we incorporate country-specific linear time trends to account for the possibility that electrification follows different underlying trajectories across countries, independently of mobile money adoption. This specification controls for unobserved, time-varying factors that evolve smoothly within countries, such as gradual infrastructure expansion, institutional improvements, or long-term development trends that could otherwise confound our estimates. Our results remain unchanged in both cases.

Finally, we conduct two additional exercises to assess the sensitivity of our results to sample composition and potential violations of the parallel trends assumption. First, in Figure A5, we exclude countries with initially high electrification rates, for which limited scope for further expansion may mechanically attenuate treatment effects or reflect structurally different electrification dynamics. Second, we perform a

sensitivity analysis using the HonestDiD approach proposed by Rambachan and Roth (2023), which relaxes the strict parallel trends assumption and allows for bounded deviations from it. This method provides robust confidence intervals that remain valid even in the presence of moderate violations of parallel trends. As shown in Figure A6, the estimated effects remain statistically significant under a wide range of sensitivity parameters.

3.3.3 Mechanisms

We are limited in our ability to pin down mechanisms in the cross-country sample; however, we observe patterns that are consistent with demand-side channels driving the relationship between electrification and mobile money adoption. In particular, mobile money may relax household liquidity constraints by facilitating financial transactions and increasing the inflow of transfers. In Figure A7, we examine the impact of mobile money adoption on remittance inflows. The results show that the introduction of mobile money leads to an increase in personal remittances over time. This pattern is consistent with the view that mobile money reduces transaction costs, thereby enabling households to receive transfers more easily and more frequently. This finding is consistent with a large body of existing evidence showing that mobile money substantially increases remittance flows and improves households' ability to smooth consumption and finance large upfront investments (Jack et al., 2013; Munyegera and Matsumoto, 2016; Apiors and Suzuki, 2018).

Additionally, although we lack direct data on fees and taxes associated with mobile money transactions, we provide suggestive evidence on the role of transaction costs by exploiting heterogeneity in the number of mobile money operators. Figure A8 presents estimates based on samples that exclude countries with 4, 5, or 6 operators, providing an indication of how the magnitude of the effect changes when countries with a larger number of operators are excluded.[5] When excluding countries with 4, 5, or 6 mobile money operators, an interesting pattern emerges: the magnitude of the estimated effect declines more when countries with a larger number of operators are excluded (see Table A10 for the aggregated effects). In other words, the reduction in the estimated effect is more pronounced when removing countries with more competitive mobile money markets. This heterogeneity suggests that lower transaction costs or improved service quality–both more likely in more competitive markets–strengthen the relationship between mobile money adoption and electrification. These findings highlight the importance of market structure and competition in shaping the effectiveness of mobile money as a driver of electrification.

3.3.4 Concurrent policies

Although our results are consistent with mobile money adoption driving increases in electrification, the lagged timing in effects makes it important to rule out any concurrent changes in policies or markets that could produce the same patterns. We first consider whether the adoption of mobile money was associated with significant increases in the supply of electricity. Second, we check for changes in fiscal or monetary policy that closely coincide with the adoption of mobile money. Finally, we check for changes in economic growth and foreign direct investment.

[5]The estimator requires a sufficiently large number of observations to achieve convergence. While combining DiD with AIPW is robust, it is also computationally demanding and inflexible. For this reason, we cannot estimate specifications excluding countries with only 1, 2, or 3 operators.

We examine potential supply-side confounds using data from the U.S. Energy Information Administration (EIA) on installed capacity and net generation from different electricity sources (EIA, 2023). Figure A9 shows that there is no change in net generation or installed capacity of electricity in countries that adopt mobile money either immediately preceding mobile money adoption or for 12 years after adoption. The results hold for all sources of electricity: fossil fuels, hydro, solar, and wind (Figure A10). This suggests that the observed positive effect on electrification rates associated with mobile money adoption is not attributable to a concurrent expansion in power supply. This is consistent with relatively low demand for electricity from rural households even once they are connected, and with rural households partially increasing access through lower cost options like solar home systems rather than grid connections. However, solar home systems became widely available in developing countries only toward the end of our study period (by 2021). Although our results may be partially driven by complementarities between mobile money and novel technologies like pay as you go solar home systems, such effects cannot fully explain the effects we observe earlier in our sample period.

Next, we examine evidence on economic policy changes concurrent with the adoption of mobile money. Figure A11 examines four types of policy changes: government spending, external debt, and two measures of inflation. Across all four indicators, we discern no perceptible trends indicating a systematic expansionary policy posture during the period surrounding the adoption of mobile money. This absence of significant fluctuations or trends in fiscal and monetary policy measures suggests that the observed effect on electrification rates following the introduction of mobile money cannot be attributed to deliberate government interventions aimed at stimulating economic activity or increasing investment in infrastructure. This reinforces the notion that the observed improvement in electrification rates is driven by the introduction of mobile money rather than concurrent policy changes.

Even if there are no concurrent economic policy changes, countries may be more likely to adopt mobile money if they are experiencing rapid growth that generates demand for cashless systems. In Figure A12, we plot effects on GDP per capita relative to the year prior to mobile money adoption and find no significant changes. Similarly, we find no evidence that Foreign Direct Investment (FDI) net inflows change significantly during the period surrounding the introduction of mobile money. The lack of significant changes in GDP per capita and FDI net inflows provides further support for our conclusion that the observed improvement in electrification rates following the introduction of mobile money is not driven by broader economic growth or increased foreign investment.

So far, our results are broadly consistent with mobile money increasing demand for electrification; however, there is one key limitation to note. We lack direct measures of electricity infrastructure. Even though the overall supply of electricity does not respond to the introduction of mobile money, the introduction of mobile money may have led to distributional shifts toward rural areas if mobile operators required electricity infrastructure to power cell towers. As such, this first-part analysis cannot fully rule out supply-side explanations.[6] We now turn to a district-level framework covering 33 sub-Saharan African countries and restrict the sample to households located in areas already covered by the power grid. This allows us to confirm whether mobile money significantly contributes to power uptake by removing demand-side barriers to electricity adoption.

[6] Although this limits our ability to precisely identify the relevant mechanisms at play at this stage, the policy implication remains intact. If the adoption of mobile money spurs expansions in electricity infrastructure to rural areas, it still indicates important complementarities between mobile money and electrification.

4 District-level mechanisms from 33 sub-Saharan African countries

4.1 Data and descriptive statistics

In our district-level analysis, we measure power uptake using Demographic and Health Surveys (DHS) from 2003 to 2021. The DHS provides nationally representative data on a set of rich covariates that we include in our analysis.

We create a continuous measure of mobile money access in a district d and year t by interacting district-level mobile network coverage with country-level adoption of mobile money access. By combining data on network coverage from Collins Mobile Coverage Explorer with data on population density from NASA's Socioeconomic Data and Applications Center, we can compute yearly rates of network access for each district in our data (see Gao and Pesaresi, 2021). We use the same GSMA database as in our previous global framework to identify the timing of mobile money introduction in a sample of 33 sub-Saharan African countries.

The descriptive statistics for all variables used in this section are reported in Table B1 and Table B2, and the data sources are reported in Table B3 and Table B4. In Figure 5, we present map of the administrative breakdown by district (ADM2) and the geographical location of the DHS clusters between 2003 and 2021. Each cluster is made up of an average of 20 households. Red dots represent clusters that are most likely not covered by an electricity network (as no household has access to electricity). The blue dots represent clusters covered by an electricity network (because at least one household is connected to the electricity network).

Only including clusters where at least one household is connected ensures that we examine the relevant sample of clusters for identifying the causal impact of mobile money on electrification through demand-side channels, as households cannot become connected to electricity in areas where there is no supply. The high concentration and isolation of many of the unconnected clusters shown on the map provides further support for our decision to exclude these unconnected clusters (in red). Indeed, for these clusters, the barriers to electricity access seem to be high enough (i.e., lack of power supply) that we would not expect a change in electrification from the mitigation of financial frictions by mobile money at the household level.

The DHS is not a household-level panel, so we instead construct a district-level panel. In our baseline estimate, we therefore consider a sample of $N = 2{,}825$ districts in our 33 sub-Saharan African countries over the period from 2003 to 2021. However, even at the district level we do not have a balanced panel because countries do not repeat the DHS at consistent intervals. This gives us a final time dimension ranging from 1 to 7 depending on the district (with an average of 1.9). The baseline number of observations or district-years ($N \times T$) is therefore 5,381.

Figure 6 presents comparison of power uptake levels between districts that have access to mobile money compared to districts that do not. We define a district with mobile money access as a district which is part of a country with mobile money and which is covered by mobile network. In box plots, the lower and upper hinges of each box indicate the 25th and 75th percentiles of the samples, the white line in the box indicates the respective medians, and the ends of the whiskers mark the next adjacent values. The figure reveals a difference in the power uptake (%) between districts with mobile money access and districts without mobile money access. Indeed, the districts with mobile money access exhibit higher median, 25th, and 75th

Table 3: Mobile money effect on power uptake, Baseline estimates using entropy balancing at district-level

	(1)	(2)	(3)	(4)
Mobile money access	0.131***	0.074***	0.127***	0.142***
	[0.013]	[0.016]	[0.013]	[0.013]
Mobile money access × GSM network coverage rate		0.080***		
		[0.017]		
Mobile money access × 3G network coverage rate			0.067***	
			[0.015]	
Mobile money access × Mobile money tax				-0.107***
				[0.013]
Observations	5382	5382	5382	5382
R^2	0.589	0.591	0.590	0.594
DHS year FE	√	√	√	√
Covariates in the 1st and 2nd stages	√	√	√	√

Notes: The first row of the table shows the average treatment effects on the treated (ATT) obtained by weighted least squares regressions (entropy balancing), and it highlights a positive and significant effect of mobile money on power uptake. The dependent variable is the power uptake (proportion of covered households) for all regressions (1-4). It is computed by dividing the number of households actually connected to electricity by the total number of households covered by the electricity grid. The treatment variable mobile money access equals to 1 when the district is within a mobile money country and covered by any mobile network (GSM or 3G), 0 otherwise. It is computed by combining data from the GSMA database and Collins Mobile Coverage Explorer. Additional control variables include the wealth index, share of rural households, share of clean cooking fuel users, average housing size, transportation index, share of households involved in agriculture, literacy rate, GDP per capita, labor tax and contributions, electricity installed capacity, and inflation. These variables are included as controls in both the first and second stages but are not reported for brevity. Except from GDP per capita, Labor tax and contributions, Electricity installed capacity and Inflation which are measured at country level, all the remaining control variables are measured at district (ADM2) level. The observations cover Demographic and Health Surveys (DHS) in Sub-Saharan Africa over the period 2003-2021. Unreported constant is included. Robust standard errors and clustered at district-level in brackets. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

(column 6). The consistency of the results across methods based on different identifying assumptions and estimation strategies provides strong evidence that our findings are not driven by the specific weighting scheme used in the baseline specification.

4.4.3 Sensitivity to the conditional independence assumption (CIA)

As with the alternative matching and weighting-based approaches, entropy balancing also assumes the unconfoundedness assumption or CIA. If the CIA does not hold up, it would imply that after conditioning on the observed covariates, there are still unobserved confounding factors that influence both the treatment assignment and the potential outcomes. To explore whether our estimates suffer from endogeneity, we first opt for the within fixed effects estimator to control for time-invariant heterogeneity across districts. District fixed effects allow us to isolate the causal effect of mobile money on power uptake by removing the potential bias that arises from unobserved heterogeneity.

In column 1 (Table B10), the effect found with the FE is also positive and significant. Like that found with the OLS (recalled in column 2), it represents less than half of our baseline effect (column 4). However, as with the OLS, when we correct for the potential endogeneity due to selection bias (see IV-FE, column 3), we find an effect which is closed to our baseline effect (column 4).

5 Conclusion

Taken together, our findings indicate a strong and positive relationship between mobile money adoption and power uptake. The effect is more pronounced in rural areas, leading to significant reductions in the urban-rural electrification gap. We find no evidence that the relationship is driven by expanded electricity supply, concurrent macroeconomic policy shifts, or changes in economic growth or foreign investment. Instead, the patterns between mobile money adoption and electrification resemble the relationship between mobile money adoption and remittance flows, potentially suggesting that the relationship is driven by demand-side factors. As highlighted in our conceptual framework, demand-side factors could include formal or informal savings and loans to overcome upfront costs for electrification, increases in income from remittance flows or new opportunities for income generation, or ease of payment.

Our results are consistent with the idea that mobile money increases demand for electrification. However, there is one key limitation to note in the section on global evidence. We lack direct measures of electricity infrastructure. Although the overall supply of electricity does not respond to the introduction of mobile money, mobile money may have led to distributional shifts toward rural areas if mobile operators required electricity infrastructure to power cell towers. Therefore, we cannot fully rule out supply-side explanations in our global analysis. We therefore turn to our district-level analysis, which focuses only on covered households and thereby allows us to fully rule out supply-side explanations. Using multiple identification strategies and estimators, we consistently find that access to mobile money increases household electrification by around 24% in our sample of districts in 33 countries across sub-Saharan Africa.

Our findings suggest that policymakers and development agencies may be able to leverage complementarities between mobile money and electrification when endeavoring to increase power uptake in developing countries. Strategies aimed at fostering mobile money adoption and at increasing the quality of mobile money services, such as regulatory support, incentives for private sector investment, and competition in the

Journal of Development Economics 176 (2025) 103478

Contents lists available at ScienceDirect

Journal of Development Economics

journal homepage: www.elsevier.com/locate/devec

Short Communication

Weather shocks, infant mortality, and adaptation: Experimental evidence from Uganda☆

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A B S T R A C T

Climate change is increasing the intensity of extreme weather events. Health is a primary channel through which climate change affects welfare. Yet, estimates of the mitigating effects of health system strengthening are largely missing. We combine data from a randomized trial inducing variation in healthcare access with naturally-occurring variation in growing-season precipitation to study the adaptive impact of community healthcare in a low-income country setting. The risk of infant death increases following low growing-season rainfall, but access to community healthcare reduces this risk by 46 %. Using our estimates coupled with projections from climatological models implies even larger potential adaptive effects.

1. Introduction

Climate change is increasing the intensity and frequency of extreme weather events, and such events are disproportionately affecting countries and populations already facing poverty and malnutrition (IPCC, 2023). For these countries and populations in particular, health is a primary channel through which climate change affects human welfare. Droughts are identified as important exposure pathways, posing multiple risks to health (IPCC, 2014). Many of these risks are indirect and affect the circumstances of living through, for example, water shortages and lower crop productivity and income. The nutritional channel – the best-recognized health impact of drought – is particularly important in Sub-Saharan Africa, where the majority of the poor rely on rain-fed agriculture for their livelihoods (Stanke et al., 2013).

The factors linking extreme weather events with increased health risks operate within the context of existing infrastructure and health service provision. Several studies have investigated mortality-mitigating channels, including exploiting quasi-experimental evidence on trade openness (Burgess and Donaldson, 2010), credit and liquidity access (Burgess et al., 2017), and water and sanitation infrastructure (Rocha and Soares, 2015; Lin et al., 2021; Ponnusamy, 2022). Yet, randomized controlled trial-based estimates of effective climate change adaptation policies in the context of health are missing (Watts et al., 2021).

In many developing countries, community health worker (CHW) programs constitute the main strategy to extend primary healthcare from facilities to underserved rural communities (Singh and Sachs, 2013). Systematic reviews of existing studies show that CHW programs can be impactful in promoting positive health behavior and in providing basic curative and health services (Bhutta et al., 2010; Haines et al., 2007; Lewin et al., 2010). Two proof-of-principle studies document large reductions in neonatal mortality (Kumar et al., 2008; Baqui et al., 2008). A study in Uganda, where CHWs were financially rewarded for providing integrated community case management (iCCM), maternal, newborn, and child health services (MNCH), and basic treatment and preventive services, reveals similar effects (Björkman Nyqvist et al., 2019).

In this paper, we combine data from the previously documented trial in Uganda – inducing spatial variation across localities in community health program coverage – with naturally-occurring variation in growing-season precipitation to assess whether adaptive investments in the form of community healthcare provision can help build climate resilience in a low-income country setting. We show that following a

☆ We thank Sam Asher, Tessa Bold, Mattia Fracchia, Seema Jayachandran, Torsten Persson and seminar participants at IIES, Stanford FSE, University of Milan, EUI, CESifo, and NOVAFRICA for helpful comments and suggestions. Financial support from the Children Investment Fund Foundation, Riksbankens Jubileumsfond (M16: 0345:1), Handelsbankens Forskningsstiftelser (P20-0242), and Mistra Foundation is greatly appreciated. The paper utilizes data from a randomized trial. The trial was registered in the American Economic Association's registry for randomized controlled trials (AEARCTR-0000530).

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https://doi.org/10.1016/j.jdeveco.2025.103478
Received 23 October 2024; Received in revised form 11 February 2025; Accepted 17 February 2025
Available online 22 March 2025
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rainfall deficit season, infant mortality increased in villages that did not host the community health program, while there is no significant relationship between mortality and rainfall in villages randomly assigned to the program. At the threshold for a moderate drought, infants in the treatment group had a 46 % lower likelihood of death than those in the control group (95 % CI: 27 %–61 %). The findings remain robust across alternative specifications and hold when examining neonatal and under-5 mortality. Although our data does not allow us to identify specific mechanisms, we provide evidence that CHWs engage more frequently with households—particularly those with sick children—after a season with poor rainfall, when the risk of mortality is higher.

The broader benefits of CHW programs depend on the expected frequency and intensity of adverse weather events. While climate models consistently predict increasing drought intensity in the coming decades, they diverge on frequency projections. We use our estimates coupled with projections from a range of models to illustrate the adaptation benefits of CHW adoption. Under models forecasting more frequent droughts, our results suggest that, depending on the realized emission pathway, CHW adoption could reduce all-season infant mortality by between 52 % and 65 %.

Our work contributes to the growing literature on the mitigation of weather-related mortality.[1] We expand on that literature by providing the first randomized controlled trial-based estimates of an effective adaptation policy for child survival in the context of climate change.[2] The design of our study parallels recent work by Gunnsteinsson et al. (2022), who used data from an RCT on at-birth vitamin A supplementation in Bangladesh and exploit the variation in health risk caused by a tornado that struck some study locations during the trial.

Our work also relates to a large body of work on the seasonal nature of income streams and infectious disease-related mortality, particularly in poor rural contexts.[3] Several studies have documented how low rainfall (droughts) around birth affects child nutrition, health, mortality, and long-run outcomes.[4] We extend this literature by assembling and utilizing detailed mortality data from 214 Ugandan villages and over 50,000 child-mortality season observations to assess the causal relationship between low growing-season precipitation and short-run child mortality risk. As in Burgess et al. (2017), we also use our estimates to investigate the potential impacts of future climate change and the mitigating effects of improved access to community health care.

Due to both data limitations and program design, our primary focus is not on identifying the specific biophysical, behavioral, or other mechanisms through which low precipitation affects child mortality, nor the precise channels access through which community health care delivers its impact. First, we lack detailed panel data to track the diverse ways in which low rainfall affects living conditions, such as threats to drinking water supplies, sanitation, the spread of vector- and water-borne diseases, and impacts on livelihoods. These rainfall-induced shocks can increase the risk of infant death both indirectly—through maternal malnutrition during pregnancy—or directly, by affecting, for example, the nutritional status of infant in early childhood. Second, as is common in similar programs, the community health program was designed to improve child health through a broad range of basic curative and preventative health services. While we provided clear evidence that health workers intensify their engagement during periods of heightened mortality risk (i.e., following a season with low rainfall), the observed reduction in mortality is likely results from the comprehensive set of health services they provide. Thus, the central question we pose—and answer—is whether, and to what extent, improved access to primary healthcare can mitigate the consequences of adverse weather events. We also use these estimates to assess the potential long-term adaptive benefits of such interventions.

The remainder of the paper is organized as follows. We introduce the details on the study setting, the research design, data, and the empirical specifications in Section II. Section III reports the results and discusses their robustness. In Section IV, we investigate potential mortality consequences of climate change and the mitigating effects of improved access to community health care. Section V concludes.

2. Study context and design

2.1. Study context

In 2007 Living Goods, a US-based NGO active in Uganda, in collaboration with BRAC Uganda, a Bangladesh-based NGO also active in Uganda, began piloting a financially incentivized community health worker program. CHWs standard tasks were to conduct home visits, educate households on essential health behaviors, provide basic medical advice, diagnose illness, and refer the more severe cases to the closest health center. Unlike the existing government-run and volunteer-based Ugandan Village Health Team program (VHT), the NGO linked monetary rewards to some performance tasks such as pregnancy registrations and newborn home visits within the first 48 h of life, and to the sale of subsidized preventive and curative health products.

The study area was predominantly rural, with small-scale, rain-fed, farming being the main source of livelihood. Consumption per adult equivalent in the study districts was US$0.91 per day in 2013 (UBOS, 2019). At baseline, community healthcare service was ineffective and in many rural villages de facto non-existent. The Ugandan VHT program received the lowest program functionality score in a WHO review implemented two years before the trial began (Bhutta et al., 2010).

As most of Uganda, villages in the study area experience two rainy seasons associated with the seasonal northward and southward migration of the Inter-Tropical Convergence Zone (Maidment et al., 2013). This bimodal rainfall pattern gives rise to the country's bimodal agricultural cycle, with the main growing seasons occurring from March to May (MAM) and September to November (SON) (Hunter et al., 2020). There is considerable spatial and time-series variation with respect to the frequency and distribution of growing-season rainfall (see Appendix A).

2.2. Study design and data

We use two sources of variation – a clustered randomized controlled trial (RCT) and a natural experiment – to assess whether adaptive investments in the form of community healthcare access can help adapt to climate change in a low-income country setting. We operationalize adaptation as mitigating the adverse impact of rainfall shocks on infant mortality.

The community health program was implemented as a three-year clustered RCT starting in 2011, with a total of 214 clusters (rural villages) spread across twelve geographical zones in ten Ugandan districts. The program assigned at least one CHW to each cluster in the treatment group (115 clusters). No CHW was assigned to the control clusters (99 clusters). Randomization was stratified by geographic zone. All households and especially households with younger children were potential recipients of visits from the CHWs. While the CHWs were recommended

[1] A related literature focuses on non-health outcomes and mitigating channels. Riley (2018), uses a difference-in-difference design to examine the effect of mobile money on smoothing income from rainfall shocks. Adhvaryu et al. (2023), as here, leverage the combination of a natural experiment (rainfall shocks) and a large-scale cluster randomized controlled trial (the PROGRESA program) to estimate the effect of early-life shocks on later-life outcomes.

[2] A related non-experimental literature examines the temperature-mortality gradient, focusing on middle- and high-income settings and primarily establishing effects for older age groups (Banerjee and Mahara, 2020; Mullins and White, 2020; Cohen and Dechezleprêtre, 2022; Helo Sarmiento, 2023).

[3] See, for example, Muñoz-Tudurí and García-Moro (2008) and Dorélien (2015).

[4] See, for example Rose (1999); Hoddinott and Kinsey (2001), Alderman et al. (2006); Jensen (2000); Maccin and Yang (2009); Currie and Vogl (2013); and Kudamatsu et al. (2012).

to focus attention on providing services to households living within their cluster, they were not prevented from selling goods or providing advice also to households outside the cluster, including control clusters. Similarly, households living outside the treatment clusters could visit CHWs in treatment clusters. Households in both treatment and control clusters could also obtain primary health care services from other actors, including private clinics, public primary health dispensaries and the VHT, and could buy medicine and other medical inputs from local drug stores. The trial's main objective was to assess the impact on child mortality of having a CHW working in the cluster. A full description of the intervention and trial design is provided in Björkman Nyqvist et al. (2019). We provide a shorter description of the intervention and trial design in Appendix B.

The primary outcome in the original trial was under-five mortality. The study, however, found similar effects for infant (under-one) and neonatal mortality. Given our focus on shorter exposure windows (six months instead of three years; see below for details), we primarily focus on infant mortality.[5] Results for under-five and neonatal mortality, reported in Appendix C, are similar. Mortality rates were measured through a cross-sectional household survey administered at the end of the trial, approximately three years after the community health program began operating in the treatment clusters. We calculate mortality rates for each cluster for six post-harvest exposure periods; i.e., for the six months following the MAM and SON rainy seasons as illustrated in Fig. 1.

2.3. Measurement

At the end of the trial, a sample household survey collected life statistics for all children under five years living in the households, as well as for all children under the age of five that had died during the trial period. We use this data to derive cluster-specific infant mortality rates for the six exposure periods over the three-year trial period (June–November following the MAM rainy season and December–May following the SON rainy season). We calculate the infant mortality rate as the number of deaths within the first year of life during each exposure period per 1000 infant-years of exposure. We compute the number of infant-years of exposure by summing up across children the number of months a given child was alive and below one year of age in a given exposure period.[6]

In the absence of detailed, high-frequency data from the study period, we also use cross-sectional data from the end-of-trial household survey to assess household interactions with health workers and treatment incidence for the two main diseases that CHWs are trained to diagnose and treat—malaria and diarrhea. Household interactions with health workers are measured using responses about visits in the 30 days preceding the interview, while malaria and diarrhea treatment data are based on a three-month recall period. Since the endline data collection occurred between September and December 2013, both the one-month and three-month recall windows fall within the same exposure season.

To derive rainfall measures in the absence of adequate gauge-based data, we rely on the satellite-based TAMSAT dataset (Maidment et al., 2017). The data – with rainfall estimates based on sensors measuring changes in cloud surface temperature to infer discharged volumes – is unique in its detailed spatial ($0.0375 \times 0.0375^{\circ}$ or 4.2×4.2 km at the equator) and temporal (daily) resolution. Validation exercises have found satisfactory accuracies of these predictions relative to ground measures (Maidment et al., 2013).

We derive a localized rainfall index by matching trial clusters with the rainfall profile of the rainfall grid cell they fall into. For each season type $s \in \{MAM, SON\}$, we calculate the cumulative amount of precipitation r^s_{jt} per cluster (j), season type (s), and year (t), over a 20-year period (December 1993–November 2013). Rainfall in our study locations displays a secular increasing trend (Fig. A3). If considering simple deviations from a long-run mean, we would identify most earlier (later) years as having 'unusually low (high)' rain, potentially contrasting with rainfall patterns as experienced by households. We address this issue by detrending the rainfall data within each study cluster through a linear regression model, and we denote the detrended rainfall as $\hat{r}^s_{jt}$.[7] Finally, we calculate the standard deviation of the detrended time series (denoted $\hat{\sigma}^s_j$) and define the rainfall index (RI) as $RI^s_{jt} = \hat{r}^s_{jt}/\hat{\sigma}^s_j$; i.e., the normalized deviation from the long-run detrended mean in local rainy season precipitation.

We interpret the rainfall index as quasi-random with regard to cluster-level outcomes. In its definition, it is similar to the Standardized Precipitation Index (SPI), a widely used measure to define (agricultural) drought conditions (Wilhite, 2000), but based on the cumulative rain during the growing season rather than on monthly rainfall. We also use the rainfall index to define rainfall deficit and surplus seasons, denoted RD^s_{jt} and RS^s_{jt}, where $RD^s_{jt} = I\left(\hat{r}^s_{jt}/\sigma^s_j \leq 0\right)$, with $I()$ an indicator function that is 1 if its argument is true and 0 otherwise. That is, a rainfall deficit season is a season with rainfall below the long-run detrended mean, and a rainfall surplus season, $RS^s_{jt} = 1 - RD^s_{jt}$, is a season with rainfall above the long-run detrended mean.

2.4. Statistical analysis

For each of the six growing seasons over the three-year trial period, we link rainfall in growing season t to infant mortality in the following six-month exposure (Fig. 1). For our cluster-level analyses, this procedure results in an effective sample size of 6 seasons x 214 clusters = 1,284 cluster-season observations. We report intention-to-treat estimates, with intention defined by the cluster of residence of the surveyed household at the onset of the study. We use three specifications. First, we report infant mortality rates conditional on assignment status (treatment and control groups) and rainfall status (rainfall deficit or surplus season), with rate ratios computed using a Poisson model, adjusting for the stratified randomized design using binary geographic zone indicators (there are twelve geographical zones in the sample). We report these estimates in Table 1 for infant mortality and Tables C2 and C3 for neonatal and under-5 mortality, respectively.

Second, we use a linear interaction specification with cluster and exposure season-specific infant mortality modelled as a function of the assignment status (T), the rainfall index (RI), and their interaction. We report the results in Table 2 for infant mortality and Table C4 for neonatal and under-5 mortality.

Finally, exploiting the same interaction specification, we estimate the risk of infant death using a logit model, controlling non-parametrically for exposure length using twelve age-bin indicators (one per month of age) and derive relative risk ratios conditional on the magnitude of the rainfall shock. The estimates for infant mortality are reported in Table 2, column (5).[8] In all specifications, we report

[5] Moreover, 78 % of the child deaths in the comparison group occurred before the child turned one year.

[6] Following the standard in the literature, we use a similar approach to estimate under-5 mortality, while for neonates we express mortality per 1000 births.

[7] The "correct" definition of the rainfall measure ultimately depends on whether one believes households to be mostly affected by deviations from the long-run mean, or instead from relatively recent rainfall patterns. We opt for the latter, even though we do not have information on households' actual beliefs and expectations. Reassuringly, our results do not depend on this choice - all results are robust to considering deviations from the long-term average (Table C7).

[8] We adopt a similar approach for under-5 deaths, controlling for exposure through 20 binary indicators that capture 3-month age intervals, from 0 until 59 months (results are reported in Table C4).

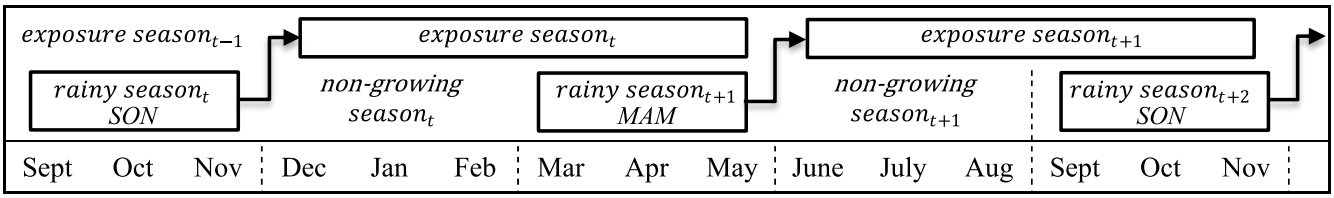


Fig. 1. Rainfall and exposure periods.
Notes. Illustration of the study design, linking precipitation in the rainy season t to mortality outcomes in the following exposure (non-growing and rainy season) season *t*. The full sample includes data for six rainy seasons (three SON, three MAM) and six exposure seasons.

Table 1
Infant mortality: summary statistics conditional on assignment status and precipitation.

Rainfall deficit season (RD)	RD = 1		RD = 0	
Assignment group	Control	Treatment	Control	Treatment
	(1)	(2)	(3)	(4)
Years of exposure to risk of death	1197	1510	1808	2033
Deaths under 1 year	83	60	77	74
Mortality rate per 1000 years of exposure	81.5	40.7	46.3	36.8
Adjusted risk ratio	0.540		0.801	
	(0.098)		(0.146)	
	[0.001]		[0.222]	

Notes. All data are from the endline sample household survey. The table displays total years of exposure to the risk of death for infants (i.e., children under 1 year) and deaths under 1 year in the sample, and mean infant mortality rate (number of infants deaths per 1,000 years of exposure to risk of death for infants) across cluster-by-exposure seasons. Rainfall Deficit (RD) indicates a growing season with rainfall below the long-run detrended mean for that location. The adjusted risk ratios; i.e., the infant mortality rate in the treatment relative to the control group, adjusting for the stratified randomized design using binary geographical zone indicators, conditional on rainfall status, are computed using a Poisson model. Standard errors, in parentheses, are clustered at the village level. There are 214 villages in the sample. *p*-values in brackets. *p*-values in brackets.

estimates adjusted only for the stratified random design. Standard errors are clustered at the rainfall grid cell-level (69 cells in the sample) to account for intra-cluster correlation across households exposed to the same rainfall shock.

3. Results

3.1. Mortality

The analysis is based on a sample of 7,018 households that lived in the same cluster throughout the trial and their 7,899 children aged less than 1 year at some point during the trial period. Baseline household characteristics, cluster characteristics, and in- and out-migration rates are evenly distributed between the two assignment groups as previously described (Björkman Nyqvist et al., 2019). We provide detailed statistics for the rainfall index and the binary rainfall deficit and surplus seasons across treatment and control clusters in Table A1, while Table C1 reports the summary statistics for all variables used in the analysis.

In Table 1 we begin by splitting the sample into exposure seasons following a rainfall deficit or a rainfall surplus season. In the control group, the infant mortality rate following a rainfall deficit season is 81.5 on average, compared to 46.3 following a rainfall surplus season. In the treatment group, mortality rates following a rainfall deficit and rainfall surplus season are broadly similar: 40.7 compared to 36.8. Adjusting only for the stratified random assignment of clusters, the mean difference corresponds to a 46 % reduction in infant mortality **following a rainfall deficit season—**an effect significant at the 1 % level (95 % confidence interval ranges from reductions of 27 %–61 %). Although mortality rates are lower in the treatment group than in the control group following a rainfall surplus season, the difference is not statistically significant, and we cannot reject the null hypothesis that mortality rates are the same in both groups. Tables C3 and C4 document similar results for under-5 and neonatal mortality, respectively. Table 2 expands on these initial results and summarizes the key findings of our analysis. Column (1) confirms that the rainfall index (*RI*) is uncorrelated with treatment status. Columns (2) and (3) show the impact of the rainfall index (*RI*) on infant mortality separately for control and treatment villages. In the control group, low growing-season rainfall leads to significantly higher infant mortality, whereas no such effect is observed in the treatment group. Column (4) reports the results of estimating an interaction specification on the full sample: rainfall impacts infant mortality only in control villages, while the linear combination of the coefficient of the rainfall index (RI) term and its interaction with the treatment indicator is close to 0 ($p = 0.311$). This is further confirmed in column (5), where we run the analysis at the child-by-season level and report the results of estimating a logit specification. Following standard practice, we report the adjusted risk ratio: a ratio greater than one indicates a higher likelihood of infant death as the variable increases, while a ratio smaller than one indicates a lower likelihood.

Panel A of Fig. 2 illustrates the differential impact of rainfall on infant mortality across treatment groups. It shows, separately for treatment and control clusters, the relationship between the risk of infant death and the amount of rainfall (based on the estimation used in column (5) of Table 2) for different values of the rainfall index. While there is a clear negative relationship between rainfall and mortality in control clusters, this relationship disappears in treatment clusters.[9] Panel B visualizes this comparison by showing that, as rainfall decreases, the relative risk of infant deaths between treatment and control groups keeps decreasing. At the threshold for a moderate drought ($RI = -0.8$), infants in the treatment group are 47 % less likely to die than infants in the control group (adjusted risk ratio is 0.53, 95 % CI: 0.39–0.73).[10]

3.2. CHW activity

CHWs deliver a broad range of preventive, promotive, and basic curative health services. However, due to the lack of detailed, high-frequency data from the study period, we cannot fully disentangle their activities to identify the specific mechanisms driving the observed results. The endline survey, however, offers some cross-sectional insights. Using this data, we construct five indicator variables. Two variables, based on responses covering the 30 days before the survey, capture whether the household was visited by: (i) any healthcare provider who provided health education and (ii) a program CHW. The remaining three variables, based on a three-month recall period, assess

[9] We observe similar patterns for neonatal and under-5 mortality, as reported in Figs. C1-C2.
[10] A SPI ≤ -0.8 is commonly used as a threshold for a moderate drought, following the classification in U.S. Drought Monitor (see Svoboda et al., 2002), with lower thresholds for severe, extreme, and exceptional droughts.

Table 2
Main results.

	Rainfall index (RI)	Infant mortality			Infant Death	Any health vist (30d)	CHW vist (30d)	Follow-up visit	Referral	Correct Treatment
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Rainfall index (RI)		−14.31	2.80	−14.02	0.793	0.051	0.054	−0.015	0.010	0.197
		(5.78)	(2.65)	(5.65)	(0.094)	(0.054)	(0.080)	(0.058)	(0.033)	(0.039)
		[0.017]	[0.296]	[0.016]	[0.049]	[0.345]	[0.500]	[0.800]	[0.761]	[0.000]
Treatment	−0.002			−29.92	0.614	0.064	0.148	0.059	0.003	−0.006
	(0.006)			(7.44)	(0.086)	(0.015)	(0.017)	(0.012)	(0.009)	(0.013)
	[0.730]			[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.728]	[0.633]
RI × Treatment				16.54	1.33	−0.077	−0.128	−0.051	−0.004	−0.040
				(6.10)	(0.176)	(0.017)	(0.020)	(0.014)	(0.008)	(0.010)
				[0.008]	[0.029]	[0.000]	[0.000]	[0.000]	[0.597]	[0.000]
Mean control						0.059	0.054	0.092	0.090	0.685
Zone FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Full	Control	Treatment	Full	Full	Full	Full	Full	Full	Full
Statistical model	OLS	OLS	OLS	OLS	Logit	OLS	OLS	OLS	OLS	OLS
Data structure	Panel	Panel	Panel	Panel	Panel	Cross-Section	Cross-Section	Cross-Section	Cross-Section	Cross-Section
Unit of Observation	Cluster-Season	Cluster-Season	Cluster-Season	Cluster-Season	Child-Season	Household	Household	Household	Household	Household
Observations	1284	594	690	1284	14876	7006	7018	4884	4884	4884

Notes. Rainfall Index (RI) captures the normalized deviation from the long-run detrended mean in local precipitation during the previous rainy season. *Infant mortality* in columns (2)–(4) is defined as the cluster-by-season infant (i.e., under 1-year) mortality rate. *Infant death* is an indicator variable that equals one if the child died before the age of one. The logit regression in column (5) controls for exposure length non-parametrically using 12 binary age-bin indicators. Columns (6)–(10) are based on cross-sectional data from the endline sample household survey. The outcomes in columns (6) and (7) are indicator variables that equal one if in the 30 days preceding the survey the household was visited by any health person that provided health education (6), or by a program CHW (7). The outcomes of the last three columns are indicator variables that equal one if in the household there was a child that fell sick with malaria or diarrhea in the three months preceding the survey and who was visited by any health specialist (8), referred to a health facility (9), or received the correct treatment (ORS and/or Zinc for treating diarrhea and ACT drug to treat malaria) (10). Binary geographical zone indicators are included in all specifications to adjust for the stratified randomized design. Standard errors, in parentheses, are clustered at the rainfall grid cell level. There are 69 cells in the sample. *p*-values in brackets.

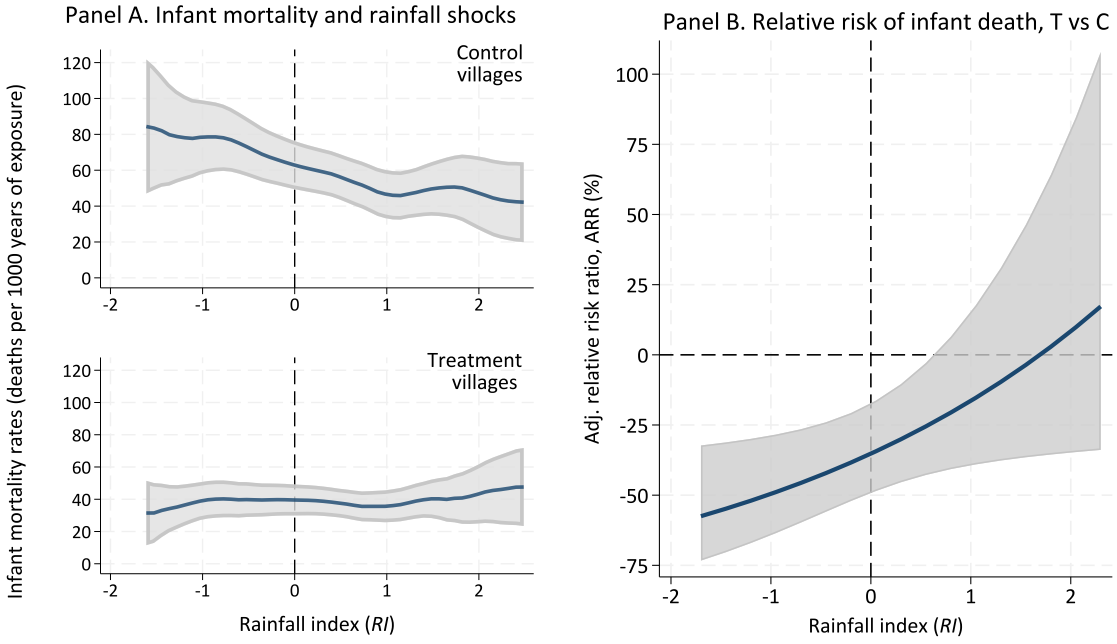


Fig. 2. Infant mortality and rainfall shocks conditional on assignment status.
Notes. Panel A displays the smoothed values of a kernel-weighted local polynomial regression of infant mortality (six months-windows) on deviations in rainfall from the long-run detrended mean (three months-windows), RI. Gray-shaded areas represent 95 percent confidence intervals for the control group (top graph) and treatment group (bottom graph). Panel B displays adjusted risk ratios from an interaction model (see Table 2, column (5)) for different values of the rainfall index, based on 14,876 child-mortality season observations. Gray shaded area represents 95 percent confidence intervals, with standard errors clustered at the rainfall grid cell level. There are 69 cells in the sample.

whether a child in the household who fell sick with malaria or diarrhea was: (iii) followed up by any healthcare provider, (iv) referred to a health facility, and (v) treated with the appropriate medication—ORS and/or Zinc for diarrhea and an ACT drug for malaria.

We estimate an interaction specification leveraging cross-sectional variation in treatment status and the rainfall index across households. Results are presented in columns (6) to (10) of Table 2. We find that under 'normal' rainfall conditions (i.e., *RI* around 0), the program increased the likelihood of household interactions with health workers both in general (column (6)) and specifically with program CHWs (column (7)). This effect is particularly evident when a child in the household fell sick with malaria or diarrhea (column (8)). More specifically, the coefficients of the interaction between the rainfall index and treatment indicators are all negative and statistically significant at the 1 % level. This indicates that households in treatment villages were significantly more likely to interact with health workers following a season of poor rainfall ($RI < 0$). The estimated effects are substantial: following a dry season with rainfall one standard deviation below the expected level, the program's impact approximately doubled.

Fig. C3 visually illustrates how the likelihood of interacting with health workers—measured by the three interaction variables—varies with the rainfall index in the treatment and control groups. Across all three variables, the graph clearly shows an increase in the intensity of interactions at low RI levels, when the risk of infant death is highest.

Table 2 also shows that the program did not affect the likelihood of referrals, regardless of the amount of rainfall in the previous season (column (9)). The negative interaction term in column (10), however, implies that the program influenced the likelihood of children receiving correct treatment for diarrhea and malaria following a season of poor rainfall.[11]

3.3. Robustness checks

As mentioned above, our results are robust to considering alternative mortality definitions, such as neonatal or under-5 mortality (see Tables C2-C5 and Figs C1 and C2). In Table C5 we show that they are also robust to the inclusion of the full set of 214 cluster fixed effects to account for any time-invariant cluster-specific characteristics. Given that the treatment was assigned at the cluster level, this specification does not allow us to estimate the treatment effect *per se*. We can, however, estimate the interaction term, exploiting rainfall variation over time. Despite the more demanding specification, the interaction coefficient remains remarkably similar to our main specification. Our results are also robust to controlling for rainfall during the previous non-growing season, as well as contemporaneous rainfall during the exposure season, both when included alone and when interacted with the treatment indicator (Table C6). This finding is especially important in light of the unavoidable imprecision in defining seasons across the country, and of the potential temporal correlation in rainfall. In particular, the inclusion of rainfall during the previous non-growing season provides a placebo check, confirming that our results are driven by the rainfall that matters most for agricultural outcomes and, thus, nutrition and health. By controlling for contemporaneous rainfall during the exposure season, we instead address the potential concern that rainfall could directly affect CHWs' ability to travel and move around the community. Finally, results are robust to using Conley standard errors to account for spatial correlation in rainfall (Table C7).

[11] In column (10) we combine the likelihood of correctly treating diarrhea and malaria. Running separate regressions for malaria and diarrhea indicates that the findings are primarily driven by correct treatment of diarrhea in periods following lower rainfall.

4. Climate change, infant deaths, and mitigation

Our results show that community health workers can effectively mitigate the effects of adverse rainfall spells on child mortality, and that these benefits increase with drought intensity. The relative benefit of rolling out a CHW program depends on the mortality cost such droughts impose, which in turn depends on their frequency and intensity. In this section, we use the estimates of the weather-mortality gradient from Section III to predict future drought-induced child mortality under various possible realizations of climate change in Uganda and illustrate the relative gain of deploying CHWs in each.

We obtain monthly precipitation projections for 2020–39 under three climate change scenarios from the 30 climate models under the Coupled Model Intercomparison Project Phase 5 (CMIP5; Taylor et al., 2012). The three scenarios represent standardized emission pathways that result in different levels of warming by the end of the century (1.8 °C, 2.7 °C, or 4.4 °C). The models incorporate these emissions into global weather simulations which also cover Uganda. We calculate projected rainfall shocks using the methodology from Section II.C.[12]

The three climate scenarios imply similar distributions of drought intensities and frequencies over the projection horizon until 2039 (Fig. 3A). Strikingly though, within each scenario, models disagree widely on whether the frequency of such droughts will increase, with the model at the 25th percentile under the 2.7 °C scenario suggesting a 40 percentage point *reduction* in drought frequency, while the model at the 75th percentile implies a 32.5 percentage point *increase*.[13] All models, however, largely agree on the *intensity* of shocks increasing, from a historical mean size of −0.8SD to −2.5SD on average across models under the 2.7 °C scenario. A pattern of fewer but more intense predicted droughts is common in Africa's tropical regions but contrasts with more frequent and more intense droughts predicted for regions away from the equator (Trisos et al., 2022).

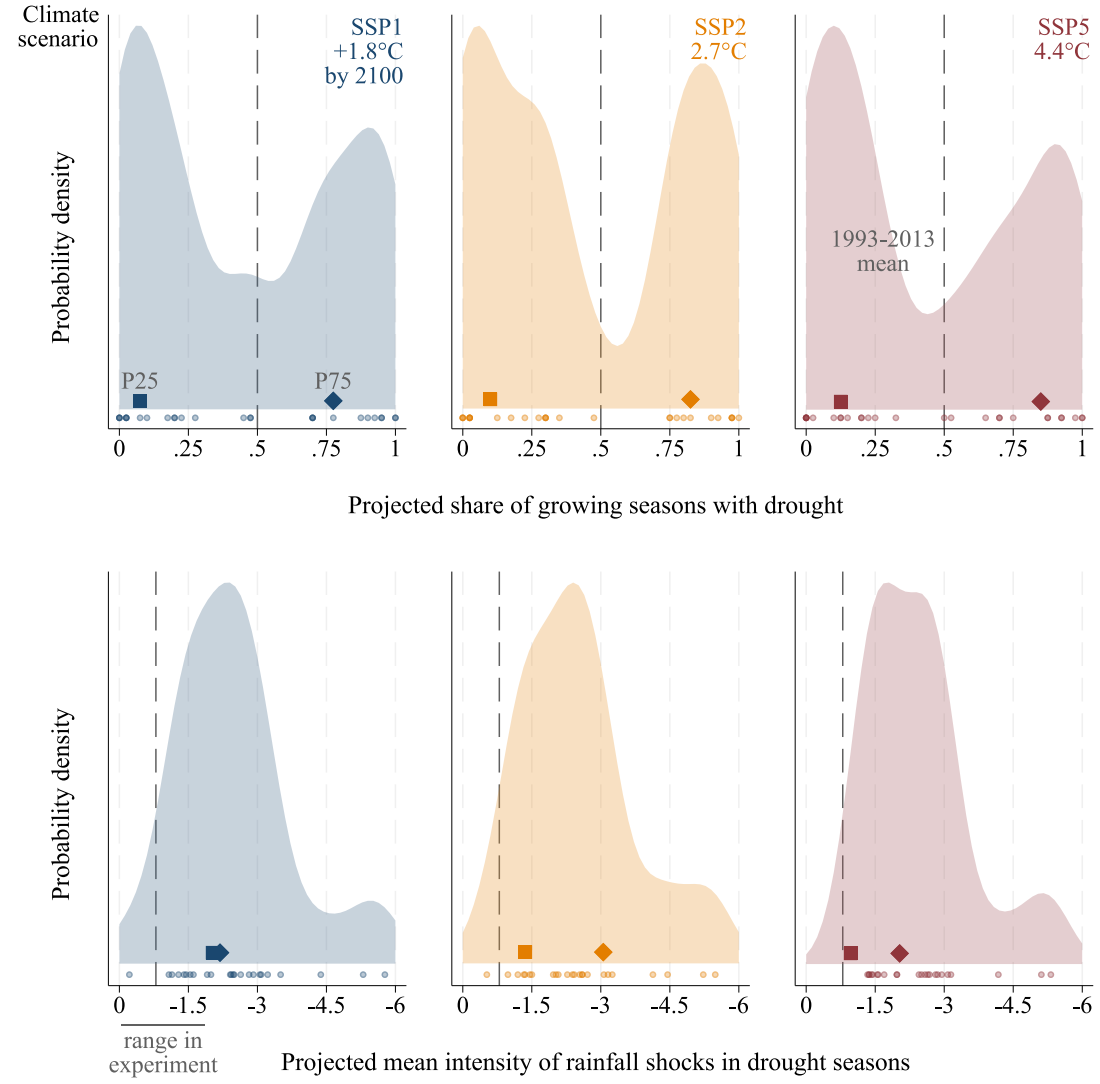
Given the high uncertainty in existing forecasts and the bimodal distribution of projected drought frequencies, in our exercise, we provide a range of estimates of the adaptive gains of CHW adoption by considering the 25th and 75th percentile models in terms of drought frequency for each scenario. Our goal is to calculate expected infant mortality for the projected rainfall patterns, taking into account their frequency and intensity, and relate it to a counterfactual of rainfall-independent mortality under a hypothetical universal CHW rollout. This requires two preparatory steps.

First, many of the projected future shocks lie outside the historical range over which we constructed the weather-mortality gradient (see Figs. 2 and 3, panel A). Specifically, in the data we observed no cluster-season episode classified as an exceptional drought (i.e., using our standardized precipitation index RI, a RI < -2), while 1.5 % of the episodes were classified as extreme droughts ($-1.9 \leq RI \leq -1.6$). To assign climate shocks of varying sizes to their expected mortality consequences, we thus need to extrapolate our estimates from Fig. 2 into the range of the forecasted shocks. For negative rainfall shocks, we do this in two ways, using either linear extrapolation according to the coefficients in Table 2, column (4), or assigning exceptional seasons ($RI < -1.9$) the lowest value for the extreme droughts from the 2011–2013 data, i.e.,

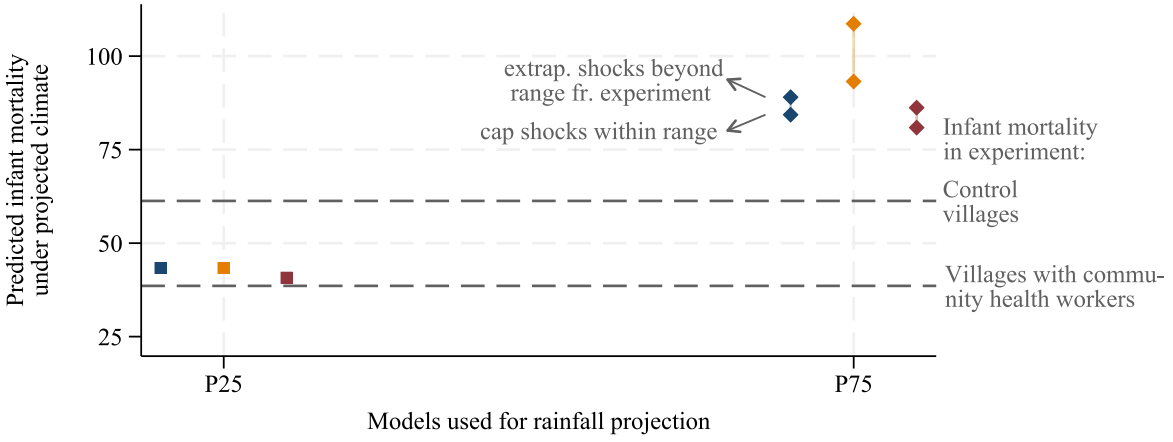
[12] In the analysis of the experiment, we use location-specific rainfall estimates. Projections at such granularity are not available. For the current exercise, we therefore use nation-wide projections, as well as a consensus record of past weather. For each model and scenario, we define the projected frequency and intensity of shocks relative to their 1993–2013 mean from the consensus weather data. Different to Section II.C, we do not detrend the rainfall projections since projections differ markedly in the degree and shape of their trends. Furthermore, rainfall trends at the national level are substantially smaller than those observed in the trial locations (Figure A5).

[13] This increase in rainfall is consistent with multi-model ensemble analysis reported in Uganda's Climate Risk Country Profile (World Bank, 2021).

(a) Projected frequency and intensity of droughts under 30 climate models, 2020-39



(b) Predicted child mortality based on experimental results and climate projections



(caption on next page)

Fig. 3. Projected rainfall and infant mortality in Uganda.
Notes. Panel A: Density plots of projected frequency and intensity of droughts during growing seasons in Uganda under 30 climate different models for three emission scenarios (1.8 °C, 2.7 °C, or 4.4 °C), indicated with three different colors. The small dots under the density plots indicate individual model outcomes. Squares and diamonds identify, respectively, the models at the 25th and 75th percentile for drought frequency. The vertical dashed lines indicate mean outcomes over the trial period. Panel B: Projected infant mortality under the three emission scenarios for the models at the 25th (squares) and 75th (diamonds) percentiles. Within each pair of symbols, the upper icon indicates projected infant mortality when we linearly extrapolate the mortality-rainfall gradient beyond the range observed during the study period (indicated in the bottom left graph of panel A). The lower symbol indicates mortality when we cap the gradient at the level of the most extreme droughts observed during the study period. This difference is negligible for the 25th percentile models. The horizontal dashed lines indicate infant mortality as observed during the trial period for treatment and control villages.

$RI = -1.9$.[14]

Second, building on our finding of smaller and not-significant mortality differences between treatment and control villages in non-deficit seasons (see Table 1), we assume that mortality in such seasons is identical for both universal CHW rollout and their complete absence, and is fixed at the average level observed in the treatment group from Table 1. We also assume that mortality under universal CHW rollout is constant during rainfall deficit seasons, fixed at the same level as during rainfall surplus seasons. These steps ensure that benefits of the hypothetical policy occur because it mitigates the negative weather-mortality gradient in locations without CHW for which we have found statistically significant evidence, and not because of the insignificant weather-mortality gradient in treatment villages.

With these preparations, we then integrate by-scenario over the projected time series of drought shocks and associated mortality levels. The exercise – summarized in Fig. 3B – illustrates that the adaptive gains from CHW deployment vary substantially across models. Should the climate change towards *less* frequent droughts, our estimates suggest an ensuing reduction in infant mortality similar to those attained via the CHW rollout during the experiment. In this case, the extrapolation method is qualitatively irrelevant as extreme drought shocks are rare. Should the climate, in contrast, change towards *more* frequent droughts, then the adaptive gains of CHW adoptions may be up to two times higher than during the period of our study. Again, this implication pertains independently of the chosen extrapolation method and across the three emission scenarios, though is more pronounced when linearly extrapolating the rainfall-mortality gradient. More specifically, under the 75th percentile model for the 2.7 °C scenario, our analysis predicts an increase in all-season infant mortality to up to 108 (under linear extrapolation; 93 under flat extrapolation) in the absence of mitigating policies. Assuming a flat weather-mortality gradient under universal CHW adoption, the policy could reduce all-season mortality by between 59 % (2.7 °C scenario, flat extrapolation) and 65 % (linear extrapolation) by 2039.[15]

These estimates underline how a warming climate, and the increased intensity of extreme weather events that may follow, may amplify the child mortality effects we estimate using the 2011–2013 mortality data, and thus speak to the importance of identifying cost-effective adaptation policies.

5. Interpretation and discussion

Climate change is generally expected to worsen the frequency, intensity, and impacts of extreme weather events. The impact is predicted to disproportionately affect countries and populations already facing poverty, malnutrition, and poor health service provision. In this paper, we provided evidence consistent with this prediction: the risk of infant death increased following adverse growing-season rainfall spells. We also showed that adaptive investments in the form of strengthened community-health service provision could significantly dampen these risks.

The results should be interpreted within the context of the study. First, the intervention was implemented across villages lacking access to effective community-health service provision at the onset of the study, and the quality of the public primary healthcare system was notoriously low (Di Giorgio et al., 2020). Furthermore, the sample population was poor. These characteristics could help explain the large magnitudes of the effects we document – and thus have bearing on the external validity of our estimates – an important limitation when making longer-run projections. At the same time, the findings highlight the inherent potential as well as feasibility to support climate resilience by strengthening community healthcare provision. The fact that the villages in the trial sample spanned all four regions of Uganda also boosts the validity of our findings.

Second, and relatedly, we estimate the effects of local rainfall shocks (as opposed to more aggregate precipitation shocks or individual shocks). While these shocks could have resulted in local prices (or relative prices) changing, these price responses are likely to have been small. Aggregate precipitation shocks may lead to larger changes in relative prices, which could both dampen or amplify the effects on income depending on households' production and consumption patterns and asset holdings. The focus on local rainfall shocks (or more aggregate shocks), also implied that local informal institutions for risk-pooling and insurance – which could have played an important role in mitigating idiosyncratic shocks to income – were likely not effective.

Third, and as discussed above, we observe few extreme events in the data. Thus, predictions about the mitigating impact of community-health program strengthening in the case of the most extreme weather events are uncertain, adding to the inherent uncertainty of predicting impacts into the future. The results, however, also suggest that for a poor rural population, less extreme deviations in growing season rainfall significantly raise the risk of infant death, and improved community-health services provision can play an important role in reducing that risk.

Data limitations prevent a direct analysis of causes of death and how the CHWs activities could have mitigated the risk of child mortality. However, since most deaths in control clusters occurred in early infancy and following low previous growing-season rainfall, it is reasonable to assume that poor maternal and child nutrition is a key driver. Undernutrition weakens immune function, increasing susceptibility to infections and heightening the lethality of childhood diseases such as pneumonia, diarrhea, and malaria (Troeger et al., 2018; Troeger et al., 2020; Rice et al., 2000). Moreover, weather-induced nutritional shocks can have particularly severe consequences in a context where undernutrition among children is widespread (Maniragaba et al., 2023), malaria is endemic, and drought conditions—especially in areas with limited access to safe drinking water and sanitation—are a major cause of diarrhea outbreaks (Dimitrova et al., 2023). Turning to how the CHW program could have mitigated these risks, CHWs activities included both actions aimed at improving nutritional outcomes and reducing the risk of severe complications of common child diseases. On the former, the CHWs actions included educating households on exclusive breastfeeding and adequate maternal and child nutrition. And as reported by

[14] Note that these extrapolations also assume that CHW efficacy in preventing deaths remains unaltered with more intense droughts. We do not consider potential negative spillovers of increased attention to children on other demographic groups.

[15] This compares to the 46 % reduction in infant mortality during rainfall deficit season documented in Table 1 and the 33.5 % reduction in mortality across all seasons documented in Table 2 (column (4)).

Björkman et al. (2019), supplemental food enriched with essential nutrients was among the most frequently purchased CHW-provided products. Regarding the latter, regular home visits were scheduled to enable early identification of cases requiring treatment and follow-up care.

In this paper, we provided to our knowledge the first causal evidence based on an RCT that improved access to basic primary health care can mitigate some of the predicted adverse consequences of climate change in a low-income setting. Adaptive health service investments could protect the most vulnerable from adverse weather events.

CRediT authorship contribution statement

Martina Björkman Nyqvist: Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Tillmann von Carnap:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Andrea Guariso:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jakob Svensson:** Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jdeveco.2025.103478.

Appendix

Appendix A. Description and summary statistics of the rainfall data

There is substantial variation in rainfall both within years (Fig. A1) and across regions of Uganda (Fig. A2). The two rainy seasons in Uganda occur from March to May (MAM) and September to November (SON) (Hunter et al., 2020). Since livelihoods are tightly linked to rain-fed agriculture, we consider rainfall shocks during these periods as potential drivers of health outcomes. Our measure of rainfall shocks considers both the general seasonal pattern as well as the locally prevailing climate.

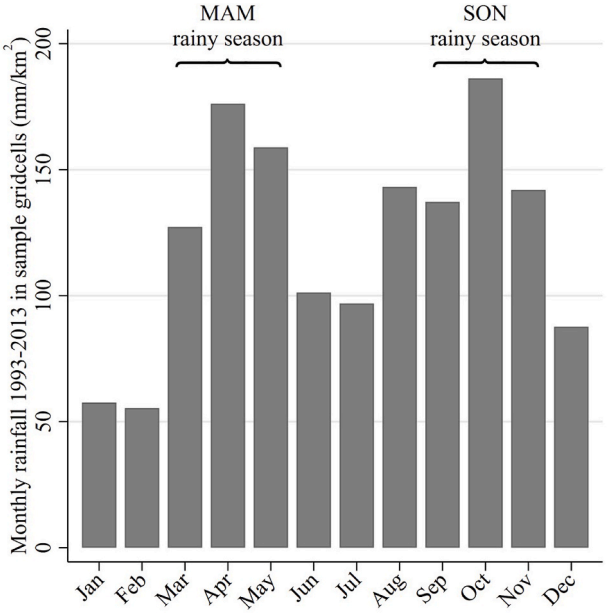


Fig. A1. Monthly rainfall in study region, 1993–2013.
Notes. Average monthly rainfall (mm/km^2) during 1993–2013 within the 69 grid cells of the TAMSAT dataset (Maidment et al., 2017) that cover the study clusters.

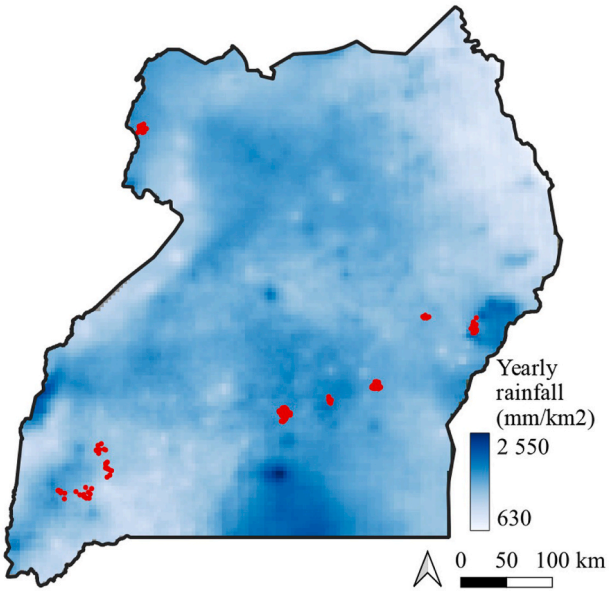


Fig. A2. Yearly rainfall average across Uganda in 2011–2013.
Notes. Average yearly rainfall (mm/km^2) during the study period 2011–2013 as recorded in the TAMSAT dataset (Maidment et al., 2017). Red dots indicate study clusters.

We base the construction of the rainfall shocks on estimates of daily rainfall from TAMSAT (Maidment et al., 2017). These estimates infer discharged rainfall volumes from remotely-sensed changes in cloud surface temperatures and are provided in raster format at a spatial resolution of 0.0375° (approx. 4.2 km at the equator). Besides its high spatial and temporal resolution, an advantage over alternative sources is that this dataset is constructed independently from the sparse ground network of weather stations in Uganda, limiting the risk of spatial correlation of rainfall estimates due to interpolation between stations. Records from existing weather stations across Africa have, however, been used for TAMSAT's extensive validation (World Meteorological Organization, 2020).

We matched each of the 214 study clusters with the rainfall time series of the raster's grid cell they fall into. Study clusters are covered by a total of 69 different grid cells. About half of the grid cells (46 %) contain only one cluster, 17 % contain two clusters, and the rest contain more than two clusters.

Our measure of rainfall shocks should be interpreted as unanticipated deviations from normal rainfall levels. To define normal rainfall levels, we consider a time window of twenty years prior to the end of the study period in 2013. Across clusters, the data displays a secular trend towards more rainfall over this time period (Fig. A3). We therefore detrended rainfall by separately estimating the following linear regression model for each cluster j and rainfall season $s \in \{MAM, SON\}$:

$$r_t^{js} = \beta_0^{js} + \beta_1^{js} t + \epsilon_t^{js},$$

where r_t^{js} is total rainfall in year t. We then estimate detrended rainfall as

$$\widehat{r}_{jt}^{\,s} = r_{jt}^{s} - \widehat{\beta}_0^{\,js} - \widehat{\beta}_1^{js} t,$$

and calculate our core shock measure as

$$\textit{Rainfall Index}_{jt}^{s} = RI_{jt}^{s} = \frac{\widehat{r}_{jt}^{\,s}}{\sigma_{\widehat{r}_j^{\,s}}}.$$

$\sigma_{\widehat{r}_j^s}$ is the shock's within-cluster and within-season standard deviation between 1993 and 2013.

Study design and participants

The study was a parallel-group, stratified cluster randomized controlled trial, embedded in the roll-out of the program. A total of 214 clusters (rural villages) took part in the trial. The clusters were located in 12 geographical zones spread over 10 districts across Uganda (see Fig. B1).

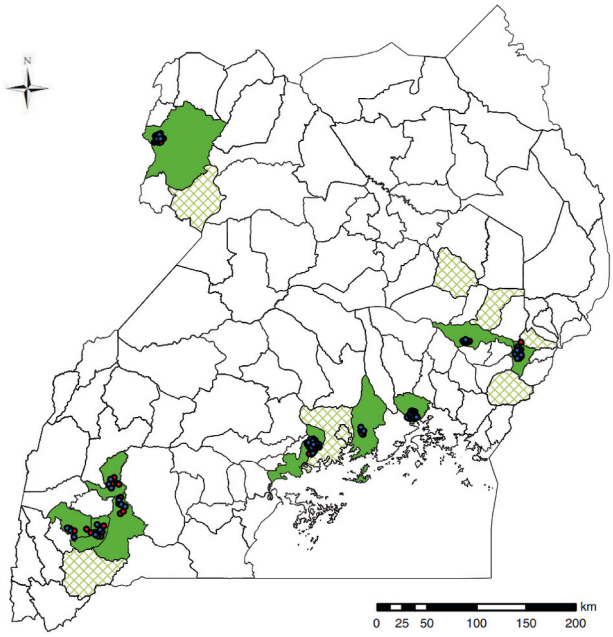


Fig. B1. Trial districts.

Notes. Green fully colored areas indicate districts that were part of the study, while cross-hatched areas indicate districts excluded from the study but in which the program was also implemented. Red and blue dots indicate, respectively, control and treatment clusters (villages) included in the study.

Within each zone, the clusters were randomly divided into a treatment group and a control group. In 11 zones out of 12, the randomization was balanced (1:1). In one zone, for operational purposes, the randomization was unbalanced (2:1). At least one program CHW was assigned to each cluster in the treatment group. No program CHW was assigned to the control clusters. The mean number of households per cluster was 237 at baseline.

The main objective of the trial was to assess the impact on under-five mortality of having a program CHW working in the cluster. The evaluation design and implementation were independent of program implementation.

The outcomes of interest were measured through a cross-sectional household survey administered between September and December 2013, approximately three years after the program CHWs began operating in the treatment clusters. Before implementing the survey, each cluster was enumerated. A random computer-generated sequence was then used to select 40 households to be surveyed in each cluster (if less than 40 eligible households were available, all were sampled). Sampled households were visited and asked for written informed consent to participate in the survey. Conditional on receiving the consent, an appointment was scheduled for the following day. The respondent was the female household head, if available at the time of the interview, or the primary female health caregiver of the household. If neither could be found, or the household refused to participate, a replacement household was chosen (this happened in 7.2 % of the cases, without any systematic difference between treatment and control clusters). Random back-checks were performed to ensure that all enumerators correctly followed the protocol. No violation was identified.

The survey was implemented by Innovations for Poverty Action (IPA) Uganda, an external organization based in Uganda and specialized in data collection. The survey teams were all composed of local staff with previous experience in data collection. Data collectors were always masked as to whether they were interviewing in a treatment or control cluster.

The program CHWs were blind to which villages were included in the study to avoid the evaluation itself affecting their behavior. As a consequence, no surveillance and monitoring system was put in place in the trial clusters, and we did not track a pre-determined set of households to avoid the CHWs focusing their efforts on the households that were tracked at the expense of those who were not. Mortality rates were calculated based on cross-sectional household survey data collected at the end of the trial, using data from households that had resided in the same cluster throughout the trial. To ensure that these households were not systematically different in the two assignment groups, we tested for differential in- and out-migration during the trial period, using pre- and post-data on the number of households residing in the clusters, and checked for balance across assignment groups using pre-trial determined observable household characteristics collected at the end of the trial period.

All households, especially those with younger children, were potential recipients of visits from the CHWs. While the CHWs were recommended to focus attention on providing services to households living within their cluster, they were not prevented from selling or providing advice also to households outside the cluster, including control clusters. Similarly, households living outside the treatment clusters could visit a CHW in a treatment cluster.

Households in both treatment and control clusters could benefit from primary health care services provided by other actors, including private clinics, public primary health dispensaries, and village health teams (a government community health worker program), and could buy medicine and other medical inputs from local drug stores.

The rollout of the program, including the trial clusters, was overseen by an advisory board including individuals with expertise in international public health and health service research as well as officials from the Uganda Ministry of Health. The trial was approved by the ethics committee of Fondazione IRCSS (D2291696), by the Harvard IRB (protocol P20141-101), by the Uganda National Council for Science and Technology (UNCST) (SS3195), and by the IRB Office of the Joint Clinical Research Center (JCRC) in Uganda. The trial was registered in the Pan African Clinical Trials Registry (#PACTR201308000601715) and the registration can be accessed online (http://www.pactr.org).

Appendix C. Additional tables and figures

Table C1
Summary statistics.

	Mean	SD	N
Panel A: Cluster-Season			
Neonatal Mortality Rate	30.20	93.78	1,284
Infant Mortality Rate	49.08	118.59	1,284
Under-5 Mortality Rate	17.30	36.55	1,284
Rainfall Deficit (RD)	0.44	0.50	1,284
Rainfall Index (RI)	0.27	0.98	1,284
Non-growing season RI	0.01	1.00	1,284
Contemporaneous RI	0.13	1.22	1,284
Panel B: Child-Season			
Neonatal Death	0.031	0.174	6,495
Infant Death	0.019	0.137	14,876
Under-5 Death	0.007	0.08	52,053
Panel C: Household (cross-section)			
Any health visit (30d)	0.103	0.304	7,006
CHW visit (30d)	0.152	0.359	7,018
Follow-up visit	0.123	0.329	4,884
Referral	0.093	0.290	4,884
Correct Treatment	0.685	0.465	4,884

Notes. In **Panel A** one observation corresponds to one season-village. *Rainfall Deficit (RD)* is an indicator variable that takes value one if there was a growing season with rainfall below the long-run detrended mean for that location. The *Rainfall Index (RI)* captures the normalized deviation from the long-run detrended mean in local precipitation during the preceding rainy season. The *Non-growing Rainfall Index (RI)* captures the normalized deviation from the long-run detrended mean in local precipitation during the preceding non-growing season. The *Contemporaneous Rainfall Index (RI)* captures the normalized deviation from the long-run detrended mean in local precipitation during the exposure season. In **Panel B** one observation corresponds to one season-child. In **Panel C** one observation corresponds to one household surveyed at *endline*. *Any health visit (30d)* is an indicator variable that takes value one if in the 30 days preceding the survey the household was visited by any health person that provided health education. *CHW visit (30d)* is an indicator variable that takes value one if in the 30 days preceding the survey the household was visited by a program CHW. *Follow-up visit* is an indicator variable that takes value one if in the household there was a child that fell sick with malaria or diarrhea in the three months preceding the survey who was visited by any health specialist. *Referral* is an indicator variable that takes value one if in the household there was a child who fell sick with malaria or diarrhea in the three months preceding the survey who was referred to a health facility. *Correct Treatment* is an indicator variable that takes value one if in the household there was a child that fell sick with malaria or diarrhea in the three months preceding the survey who received the correct treatment (ORS and/or Zinc for treating diarrhea and ACT drug to treat malaria).

Table C2
Neonatal mortality rates conditional on assignment status and rainfall.

Rainfall deficit season (RD)	RD = 1		RD = 0	
Assignment group	Control	Treatment	Control	Treatment
	(1)	(2)	(3)	(4)
Number of births	1217	1524	1758	1996
Deaths under 1 month	57	44	49	54
Mortality rate per 1000 births	45.7	27.6	27.5	24.4
Adjusted risk ratio	0.56		0.91	
(95 % CI)	(0.37–0.85)		(0.60–1.39)	
[p-value]	[0.007]		[0.676]	

Notes. All data are from the endline sample household survey. The table displays total number of births and deaths within the first month of life in the sample, and mean neonatal mortality rate (number of neonatal deaths per 1,000 births) across cluster-by-exposure seasons. Rainfall Deficit (RD) indicates a growing season with rainfall below the long-run detrended mean for that location. The adjusted risk ratios; i.e., the neonatal mortality rate in the treatment relative to the control group, adjusting for the stratified randomized design using binary geographical zone indicators, conditional on rainfall status, are computed using a Poisson model. Standard errors are clustered at the village level. There are 214 villages in the sample. *p*-values in brackets.

Table C3
Under-5 mortality rates conditional on assignment status and rainfall.

Rainfall deficit season (RD)	RD = 1		RD = 0	
Assignment group	Control	Treatment	Control	Treatment
	(1)	(2)	(3)	(4)
Years of exposure to risk of death	4302	5192	6392	7070
Deaths under 5 years	108	86	98	96
Mortality rate per 1000 years of exposure	27.4	16.4	15.7	12.7
Adjusted risk ratio	0.59		0.83	
(95 % CI)	(0.44–0.81)		(0.61–1.14)	
[p-value]	[0.001]		[0.250]	

Notes. All data are from the endline sample household survey. The table displays total years of exposure to the risk of death under 5 years and deaths under 5 year in the sample, and mean under-5 mortality rate (number of under-5 deaths per 1,000 years of exposure to risk of death for infants) across cluster-by-exposure seasons. Rainfall Deficit (RD) indicates a growing season with rainfall below the long-run detrended mean for that location. The adjusted risk ratios; i.e., the under-5 mortality rate in the treatment relative to the control group, adjusting for the stratified randomized design using binary geographical zone indicators, conditional on rainfall status, are computed using a Poisson model. Standard errors are clustered at the village level. There are 214 villages in the sample. *p*-values in brackets.

Table C4
Neonatal and under-5 mortality.

	Neonatal		Under-5	
	Mortality	Death	Mortality	Death
	(1)	(2)	(3)	(4)
RI × Treatment	7.74	1.290	3.48	1.238
	(4.47)	(0.176)	(1.73)	(0.156)
	[0.088]	[0.062]	[0.048]	[0.091]
Treatment	−14.26	0.676	−8.00	0.676
	(5.26)	(0.080)	(2.24)	(0.073)
	[0.008]	[0.001]	[0.001]	[0.000]
Rainfall index (RI)	−6.26	0.865	−3.32	0.850
	(5.26)	(0.098)	(1.48)	(0.083)
	[0.155]	[0.203]	[0.028]	[0.098]
Sample	Full	Full	Full	Full
Statistical model	OLS	Logit	OLS	Logit
Zone FE	Yes	Yes	Yes	Yes
Observations	1284	6495	1284	50084

Notes. Neonatal mortality in column (1) is defined as the cluster-by-season neonatal (i.e. within the first month of life) mortality rate. Neonatal death in column (2) is an indicator variable that takes value one if the child died within the first month of life. Under-5 mortality in column (3) is defined as the cluster-by-season under-5 (i.e. under 5 years) mortality rate. Under-5 death in column (4) is an indicator variable that takes value one if the child died before reaching the age of 5. The logit regression in column (5) controls for exposure length non-parametrically using 20 binary indicators that capture 3-month age intervals (from 0 until 59 months). Binary geographical zone indicators are included in all specifications to adjust for the stratified randomized design. Standard errors, in parentheses, are clustered at the rainfall grid cell level. There are 69 cells in the sample. *p*-values in brackets.

Table C5
Cluster fixed effects.

	Infant Mortality	Infant Death
	(1)	(2)
RI × Treatment	16.89	1.411
	(6.93)	(0.212)
	[0.018]	[0.022]
Rainfall index (RI)	−14.12	0.752
	(6.24)	(0.094)
	[0.027]	[0.023]
Sample	Full	Full
Statistical model	OLS	Logit
Cluster FE	Yes	Yes
Observations	1284	14876

Notes. The table replicates the key mortality results from Table 2 (columns (4) and (5)) of the main text, by including cluster fixed effects. The logit regression in column (2) controls for exposure length non-parametrically using 12 binary age-bin indicators. Standard errors, in parentheses, are clustered at the rainfall grid cell level. There are 69 cells in the sample. *p*-values in brackets.

Table C6
Alternative rainfall indexes.

Alternative RI:	Rainfall in previous non-growing season						Rainfall in the mortality season					
	Infant mortality	Infant Death	Any health vist (30d)	CHW vist (30d)	Follow-up visit	Correct Treatment	Infant mortality	Infant Death	Any health vist (30d)	CHW vist (30d)	Follow-up visit	Correct Treatment
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
RI × Treatment	18.62	1.336	−0.064	−0.133	−0.061	−0.026	16.51	1.337	−0.120	−0.147	−0.062	−0.045
	(7.30)	(0.200)	(0.022)	(0.027)	(0.018)	(0.015)	(6.04)	(0.176)	(0.028)	(0.032)	(0.025)	(0.023)
	[0.013]	[0.053]	[0.004]	[0.000]	[0.001]	[0.080]	[0.008]	[0.028]	[0.000]	[0.000]	[0.015]	[0.047]
Treatment	−30.51	0.614	0.035	0.165	0.082	−0.034	−30.84	0.610	0.092	0.158	0.066	−0.003
	(7.78)	(0.086)	(0.032)	(0.039)	(0.028)	(0.025)	(7.71)	(0.092)	(0.018)	(0.020)	(0.016)	(0.012)
	[0.000]	[0.000]	[0.286]	[0.000]	[0.005]	[0.184]	[0.000]	[0.001]	[0.000]	[0.000]	[0.000]	[0.811]
RI	−15.75	0.779	0.048	0.050	−0.015	0.194	−14.06	0.763	0.066	0.047	0.018	0.193
	(6.66)	(0.100)	(0.052)	(0.079)	(0.058)	(0.040)	(5.53)	(0.089)	(0.051)	(0.078)	(0.060)	(0.043)
	[0.021]	[0.052]	[0.366]	[0.532]	[0.803]	[0.000]	[0.013]	[0.020]	[0.199]	[0.550]	[0.766]	[0.000]
Alt.RI × Treatment	−4.31	0.994	−0.055	0.027	0.041	−0.052	7.12	1.055	0.053	0.024	0.014	0.007
	(7.79)	(0.170)	(0.042)	(0.060)	(0.041)	(0.043)	(5.21)	(0.124)	(0.020)	(0.022)	(0.019)	(0.022)
	[0.582]	[0.973]	[0.194]	[0.658]	[0.323]	[0.228]	[0.176]	[0.647]	[0.011]	[0.282]	[0.462]	[0.748]
Alt.RI	3.50	1.042	−0.038	−0.115	−0.013	−0.025	−8.35	0.847	−0.173	−0.187	−0.088	−0.056
	(6.09)	(0.124)	(0.056)	(0.075)	(0.064)	(0.066)	(4.20)	(0.064)	(0.039)	(0.056)	(0.051)	(0.062)
	[0.568]	[0.731]	[0.505]	[0.131]	[0.834]	[0.703]	[0.051]	[0.027]	[0.000]	[0.001]	[0.090]	[0.374]
Zone FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Full	Full	Full	Full	Full	Full	Full	Full	Full	Full	Full	Full
Statistical model	OLS	Logit	OLS	OLS	OLS	OLS	OLS	Logit	OLS	OLS	OLS	OLS
Data structure	Panel	Panel	Cross-Section	Cross-Section	Cross-Section	Cross-Section	Panel	Panel	Cross-Section	Cross-Section	Cross-Section	Cross-Section
Unit of Observation	Cluster-Season	Child-Season	Household	Household	Household	Household	Cluster-Season	Child-Season	Household	Household	Household	Household
Observations	1284	14876	7006	7018	4884	4884	1284	14876	7006	7018	4884	4884

Notes. The table replicates the key results from Table 2 in the main text, by adding to the original specification an alternative rainfall index definition. The standard Rainfall Index (RI) captured the normalized deviation from the long-run detrended mean in local precipitation during the preceding rainy season. In columns (1)–(6) the alternative rainfall index is defined as the normalized deviation from the long-run detrended mean in local precipitation during the preceding non-growing season. In columns (7)–(12) the alternative rainfall index is defined as the normalized deviation from the long-run detrended mean in local precipitation during the exposure season. Columns (2) and (8) reports risk ratios and the underlying logit regressions control for exposure length non-parametrically using 12 binary age-bin indicators. Binary geographical zone indicators are included in all specifications to adjust for the stratified randomized design. Standard errors, in parentheses, are clustered at the rainfall grid cell level. There are 69 cells in the sample. *p*-values in brackets.

Table C7
Alternative RI definition.

	Rainfall index (RI)	Infant mortality			Infant Death	Any health vist (30d)	CHW vist (30d)	Follow-up visit	Referral	Correct Treatment
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Rainfall index (RI)		−12.30	3.29	−11.65	0.845	0.056	0.142	−0.027	0.031	0.096
		(6.17)	(3.06)	(5.99)	(0.098)	(0.046)	(0.083)	(0.052)	(0.038)	(0.054)
		[0.052]	[0.2689	[0.056]	[0.148]	[0.226]	[0.091]	[0.600]	[0.416]	[0.083]
Treatment	−0.003			−35.94	0.559	0.116	0.241	0.097	0.005	0.020
	(0.007)			(9.14)	(0.086)	(0.023)	(0.027)	(0.019)	(0.010)	(0.017)
	[0.665]			[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.616]	[0.246]
RI × Treatment				14.27	1.28	−0.083	−0.158	−0.065	−0.005	−0.040
				(6.41)	(0.162)	(0.021)	(0.022)	(0.017)	(0.009)	(0.013)
				[0.029]	[0.054]	[0.000]	[0.000]	[0.000]	[0.628]	[0.002]
Mean control						0.059	0.054	0.092	0.090	0.685
Zone FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Full	Control	Treatment	Full	Full	Full	Full	Full	Full	Full
Statistical model	OLS	OLS	OLS	OLS	Logit	OLS	OLS	OLS	OLS	OLS
Data structure	Panel	Panel	Panel	Panel	Panel	Cross-Section	Cross-Section	Cross-Section	Cross-Section	Cross-Section
Unit of Observation	Cluster-Season	Cluster-Season	Cluster-Season	Cluster-Season	Child-Season	Household	Household	Household	Household	Household
Observations	1284	594	690	1284	14876	7006	7018	4884	4884	4884

*Notes.*The table replicates the key results from Table 2 in the main text, defining the Rainfall Index (RI) in terms of normalized deviation from the long-run mean (as opposed to the long-run *detrended* mean) in local precipitation during the previous rainy season. Infant mortality in columns (2)–(4) is defined as the cluster-by-season infant (i.e., under 1-year) mortality rate. Infant death is an indicator variable that equals one if the child died before the age of one. The logit regression in column (5) controls for exposure length non-parametrically using 12 binary age-bin indicators. Columns (6)–(10) are based on cross-sectional data from the endline sample household survey. The outcomes in columns (6) and (7) are indicator variables that equal one if in the 30 days preceding the survey the household was visited by any health person that provided health education (6), or by a program CHW (7). The outcomes of the last three columns are indicator variables that equal one if in the household there was a child that fell sick with malaria or diarrhea in the three months preceding the survey and who was visited by any health specialist (8), referred to a health facility (9), or received the correct treatment (ORS and/or Zinc for treating diarrhea and ACT drug to treat malaria) (10). Binary geographical zone indicators are included in all specifications to adjust for the stratified randomized design. Standard errors, in parentheses, are clustered at the rainfall grid cell level. There are 69 cells in the sample. *p*-values in brackets.

Table C8
Conley standard errors.

	Infant mortality	Any health vist (30d)	CHW vist (30d)	Follow-up visit	Referral	Correct Treatment
	(1)	(2)	(3)	(4)	(5)	(6)
Rainfall index (RI)	−14.02	0.051	0.054	−0.015	0.010	0.197
	(0.016)	(0.345)	(0.500)	(0.800)	(0.761)	(0.000)
	[0.003]	[0.000]	[0.246]	[0.713]	[0.581]	[0.000]
	{0.005}	{0.073}	{0.387}	{0.781}	{0.433}	{0.000}
Treatment	−29.92	0.064	0.148	0.059	0.003	−0.006
	(0.000)	(0.000)	(0.000)	(0.000)	(0.728)	(0.633)
	[0.000]	[0.009]	[0.000]	[0.000]	[0.560]	[0.514]
	{0.000}	{0.003}	{0.000}	{0.000}	{0.549}	{0.497}
RI × Treatment	16.54	−0.077	−0.128	−0.051	−0.004	−0.040
	(0.008)	(0.000)	(0.000)	(0.000)	(0.597)	(0.000)
	[0.000]	[0.001]	[0.000]	[0.000]	[0.438]	[0.000]
	{0.000}	{0.000}	{0.000}	{0.000}	{0.407}	{0.000}
Zone FE	Yes	Yes	Yes	Yes	Yes	Yes
Data structure	Panel	Cross-Section	Cross-Section	Cross-Section	Cross-Section	Cross-Section
Unit of Observation	Cluster-Season	Household	Household	Household	Household	Household
Observations	1284	7006	7018	4884	4884	4884

Notes. The table replicates the key results from Table 2 in the main text and reports three sets of *p*-values: i) original *p*-value from the main paper, associated with clustering standard errors at the rainfall grid cell level (in parentheses); ii) *p*-values associated with Conley standard errors with 50 km distance cutoff (in square brackets); iii) *p*-values associated with Conley standard errors with 100 km distance cutoff (in curly brackets).

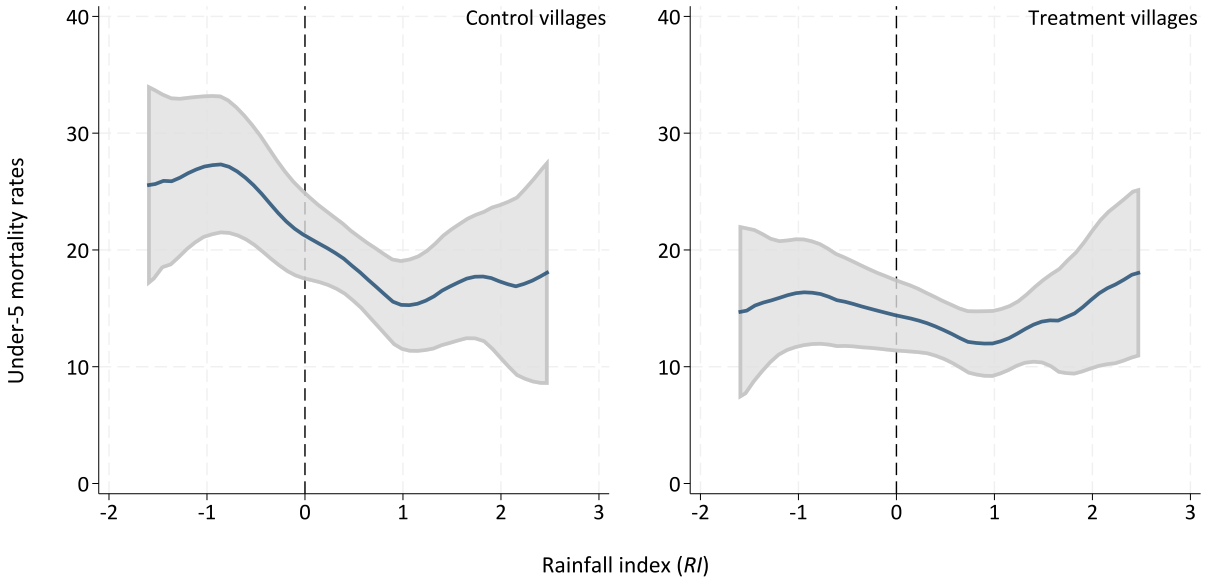


Fig. C1. Under-5 mortality and rainfall shocks conditional on assignment status.
Notes. Smoothed values from a kernel-weighted local polynomial regression of under-5 mortality (six months windows) on deviations in rainfall from long run detrended mean (three months windows) based on 50,084 child-mortality season observations. Gray shaded area represents 95 percent confidence intervals.

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World Development Vol. 67, pp. 267–280, 2015
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http://dx.doi.org/10.1016/j.worlddev.2014.10.022

Social Preferences and Agricultural Innovation: An Experimental Case Study from Ethiopia

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Summary. — We run an experiment in Ethiopia where farmers can use their own money to decrease the money of others (money burning). The data support the prediction from an inequality aversion model based on absolute income differences; but there is no support for an inequality aversion model based on comparison with mean payoff of others. Experimentally measured money burning on the village level is negatively correlated to real-life agricultural innovations. This result is robust even when data from another independent survey than the current research are used. This underscores the importance of social preferences in agricultural innovations in developing countries.
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Key words — social preferences, money burning, agricultural innovations, Ethiopia

1. INTRODUCTION

Research on social preferences in economics has by now amply demonstrated that the welfare of individuals is affected not only by the goods and services they consume but also by the position and actions of others with whom they compare themselves. Most of the literature on social preference focuses on pro-social behavior where cooperation and other regarding behavior lead to better socially beneficial outcomes than predicted by the standard economic model. For example, when punishment is introduced in public good games, people are willing to punish those that free-ride, even at their own expenses, and are reluctant to punish those that cooperate (Ertan, Page, & Putterman, 2009). In trust games, players give substantial proportion of their money to others with the expectation that they will get it back even though they have no control on the decision of the recipients (Cox, 2004). The literature on reciprocity emphasizes mainly its role in strengthening pro-social behavior. One of the celebrated books in this area, Henrich *et al.* (2004), summarizes results of many experimental games and ethnographic studies from fifteen small-scale societies in different parts of the world. The main thrust of this strand of literature is toward understanding how pro-social behavior helps overcome social dilemmas and improve social outcomes. [1]

In contrast, a growing body of literature focuses on the negative aspects of social preferences where the destruction of potential surplus (value) is emphasized. Earlier work by Kirchsteiger (1994) suggested that envy is an equally plausible underlying motive as fairness for rejection of offers in the ultimatum game. Experimental work suggests that people are willing to devote their resources to decrease the welfare of better-off people (Zizzo, 2003; Zizzo & Oswald, 2001) – specifically, they are willing to destroy ('burn') other people's earnings at a cost to themselves. Laboratory experiments also show that subjects are willing to harm for little reason or no self-interest (Abbink & Herrmann, 2009, 2011). Experiments conducted in India show that spiteful preferences – the desire to reduce another's material payoff for the mere purpose of increasing one's relative payoff – are widespread (Fehr, Hoff, & Kshetramade, 2008).

In contrast to the papers cited above that used experimental games, research using survey methods also looked at positional concerns – whether the relative position of people matters. Some of the papers argue that positional concerns become important only at higher levels of income; they argue people in low-income countries are mainly concerned about their absolute positions rather than their relative standing compared to others (Akay, Martinsson, & Medhin, 2012; Clark, Frijters, & Shields, 2008; Frey & Stutzer, 2002). In contrast, others found just the opposite. For example, Corazzini, Esposito, and Majorano (2011) found positional concerns in fact are higher in developing compared to rich countries. Other studies in developing countries also show relative positions matter. A recent study on India (Fontaine & Yamada,

* We wish to thank the Economic and Social Research Council (RES-000-22-2840) and the University of East Anglia for financial support, the Ethiopian Economic Association and the University of Addis Ababa for their help in setting up the experimental work. We also thank Alistair Munro, Arjan Verschoor, and participants of the following for their comments: a workshop in Addis Ababa, three presentations at the University of East Anglia, the DIME Conference on Experimental Methods on Innovation: Broadening and Challenging Existing Knowledge (Jena, Germany, November 2009), the annual conference of the Centre for the Study of African Economies (CSAE; Oxford, March 2010), Economic Science Association conference (Copenhagen, July 2010), seminars in Kent at Canterbury and in Nottingham (November 2010), the SIRE Young Researchers Forum (Edinburgh, March 2011), the Royal Economic Society conference (Royal Holloway, April 2011), Symposium on Economic Experiments in Developing Countries (SEEDEC) at the University of California, Berkeley (December 2011) and a presentation at the University of Manchester (February 2012). We are also very grateful to the editor of this journal and the anonymous reviewers for very useful comments; the paper has significantly improved because of their feedback. We also thank Axel Sontag for assistance in data analysis. The usual disclaimer applies. The experimental data are available on the UK Data Archive. Final revision accepted: October 21, 2014.

2014) found three interesting results that support positional concerns are important in developing countries. First, within-caste comparisons reduce well-being; expenditure by others from the same caste triggers stronger envy than providing a positive signal about one's future prospects. Second, between-caste comparisons have a stronger effect than within-caste comparisons. Third, high castes' economic successes are detrimental to low castes' well-being but the reverse is not true. In a different context, a study from Jordan (El-Said & Harrigan, 2009) found strong envy between the Jordanian and Palestinian (immigrant) communities – while Jordanians envy the Palestinians' dominance in the higher wage private sector, Palestinians envy the Jordanians because of the latters' dominance in the public sector. Theesfeld (2004) discusses how distrust and envy creates a formidable constraint for collective action required to maintain irrigation in Bulgaria.

This paper mainly falls within the second strand of literature that focuses on the negative effects of social preferences. To capture this aspect, we employ a money burning experimental design (Zizzo, 2003; Zizzo & Oswald, 2001) in rural villages of Ethiopia, with additional sessions with university students in an urban area. To our knowledge, this is the first study that uses money burning games in a rural setting of a developing country. Our money burning game has two parts: in the first, an unequal distribution of resources is induced by varying initial endowments after which participants play a lottery. In the second round, people are given a chance to use their own money to decrease ('burn') others' money. In addition to exploring the existence and variations of money burning behavior, one of the main objectives of this paper is to understand the underlying motives. Our experimental design provides data for testing the deeper motives for money burning behavior. In particular, we test whether inequality aversion and/or reciprocity/retaliation play a role. There is evidence that players burn money due to inequality aversion based on absolute income differences as suggested by the Fehr and Schmidt (1999) model.

The money burning behavior of participants in the game is likely a reflection of similar behavior in reality. Our particular interest in this regard is to link our experimental results with real-life agricultural innovations. There are many anecdotal evidences in a number of countries of sabotaging behavior which targets better-off individuals. There is at least a potential case that this likely discourages entrepreneurship (Schoeck, 1966). Mui (1995) gives examples from reforming East European countries and China of how sabotaging may have constrained the emergence of entrepreneurs. Caplan (2005) discusses the 'cargo system' in rural Latin America where successful individuals expected to hold offices are required to self-fund and that this may result in an informal tax as high as 80%. He notes how this may discourage innovation and growth. Earlier sociological and psychological work on Ethiopia also emphasizes the pressure toward conformity and the zero-sum nature of social interactions in the country (Korten, 1972; Levine, 1965, 1974).

Agricultural innovations are complex processes that are affected by many factors – there is voluminous theoretical and empirical literature on agricultural innovations in developing countries. Feder and Umali (1993) and Sunding and Zilberman (2001) are good reviews of the literature on agricultural innovations. Ward and Singh (2014) is a recent paper that relates experimental games with agricultural innovations in a developing country.[2] Providing an exhaustive review of the literature is beyond this paper but highlighting some recent findings on agricultural innovations on Ethiopia is instructive.

Most farmers in rural Ethiopia live in a highly unpredictable environment facing such significant environmental shocks like draught; smoothing consumption across time is a real challenge. Dercon and Christiaensen (2011) showed lower consumption due to harvest failure is an important constraint in fertilizer adoption, implying that consumption smoothing is an important determinant of innovation. The influence of social networks in encouraging the spread of information and knowledge and consequently enhancing innovations has been emphasized in the recent literature. Krishnan and Patnam (2013) compared learning from government extension agents and from neighbors in the adoption of fertilizer and improved seeds in Ethiopia. They found that, while the initial impact of extension agents was high, the effect wore off after some time, in contrast to learning from neighbors underlining the importance of social networks. Abebe, Bijman, Pascucci, and Omta (2013) emphasized the market-related quality attributes in the choice of new crops such as improved variety potatoes in Ethiopia – the spread of improved variety potato was constrained by the preference of people for the taste of the local variety.

The growing empirical literature increasingly shows the complex nature of agricultural innovations. Yet the role of social preferences in agricultural innovations is still not well-understood. This paper contributes toward this by linking behavior observed in experimental games with real-life agricultural innovations. For that purpose, this project deliberately uses subjects that were previously covered by a panel household survey to utilize already available data on agricultural innovations. Multi-level mixed effects models that control for village- and session-level random effects are used to identify the link between agricultural innovations and money burning behavior observed in the experimental games. The empirical results show a robust negative correlation between social money burning and agricultural innovations – observed agricultural innovations as captured by an independent previous survey are lower in communities with high rates of money burning. We further use information on three innovations – fertilizer, improved seeds, and rain water harvesting – on which the current research collected data and also found robust and negative correlation between social money burning rates and agricultural innovations. These results imply that the money burning behavior captured by the game in the laboratory most likely captures an unobservable social preference that is inimical to real-life agricultural innovations. The link created between the behavior of participants in the game and their real-life agricultural innovations is the other important contribution of this paper. This is supported by qualitative data coming from sociological reports prepared as part of our project, for example with one of the farmers surveyed reporting that "using better technology might be good in terms of increasing yields. But it also increases the number of enemies one might have. You will be targeted by enemies including wild animals and those who possess the power of the evil eye; they will affect your cattle's fertility as well as the fertility of the soil permanently" (Dessalegn, 2009).

The rest of the paper is organized as follows. Section 2 describes the experimental design and the inequality aversion models predictions. While results are discussed in Section 3, Section 4 provides the conclusions.

2. EXPERIMENTAL DESIGN AND INEQUALITY AVERSION MODELS PREDICTIONS

(a) *Design*

Thirty individuals participate in a session of the experimental game. At the start, players are randomly given large (Birr 15) or small (Birr 7) amounts of money to induce inequality.[3]

We shall call the first high- and the second low-income players. Players know whether they are high- or low-income players. Then players can use any amount of their initial endowment to buy a more than actuarially fair lottery with a 50% chance of winning three times the amount invested. The thirty players are divided into five groups with six players in each group, equally divided into high- and low-income players. Anonymity within each group is strictly maintained, i.e., even though the thirty participants in a session can see each other, with whom they are matched in a group is kept unknown.

After the lottery game, players are informed of what amounts of money the other five members of their group initially received, saved, and how much they have won from the lottery. All six members of a group are then asked how much of the money of others in their group they would like to 'burn'; each player makes five money burning decisions related to each of the other five members of the group. Players cannot burn their own money. They have to pay, however, for burning the money of others; the price of money burning is one tenth of the amount to burn (for example, a player has to pay Birr 1 to decrease another's money by Birr 10). The money burning is separately done for 'safe' (that is, un-invested) money and lottery winnings in order to capture potential heterogeneity.

After eliciting the money burning decisions, a random dictator design is used to determine the actual money to be burnt. Even though the amounts the six players want to burn from the money of the other five are recorded, only the choice of one is implemented after random selection.[4] Subjects learn about the outcome of the game before moving to the next one.

The game is repeated three times (three *stages*). Two sessions of 30 subjects were run in each of the four rural villages (a total of 240 players). In each stage subjects were randomly matched with their group members (*variable groups*). Two variable group sessions were also conducted at Addis Ababa University with 60 students. Additional two sessions in which the same subjects were matched in the three stages (*fixed groups*) were run with 60 students at the university.[5]

At the end of the experiment, participants went home with all the money accumulated over the three stages plus a participation fee of Birr 40. The mean payment, including the participation fees, was around Birr 80, i.e., approximately U.S. $10 or around 4 days of wages for unskilled labor.

(b) *Inequality aversion model predictions*

Economists increasingly recognize that people care about their relative position in addition to their own payoff. The models of Fehr and Schmidt (1999) and Bolton and Ockenfels (2000) are two popular models that formalize this concern specifically concentrating on inequality aversion. In the inequality aversion model of Fehr and Schmidt (1999), players dislike inequality whether it is advantageous or disadvantageous to them; richer people feel 'guilt' and poorer people suffer from 'envy' (this is the reason why the Fehr–Schmidt model is sometimes referred to as the 'guilt/envy' model). Even though people can feel 'guilt' or 'envy' depending on their relative position, the latter motive is stronger than the former – being comparatively rich is better than being comparatively poor. The Fehr–Schmidt utility function has the following form:

$$U_i(x) = x_i - \alpha_i \frac{1}{n-1}\sum_{j\neq i}\max\{x_j - x_i, 0\} - \beta_i \frac{1}{n-1}\sum_{j\neq i}\max\{x_i - x_j, 0\} \quad (1)$$

Here x_i denotes monetary payoff of player i and n represents the number of players. The second term in the utility function measures the loss of utility from disadvantageous inequality ('envy') and the third from advantageous inequality ('guilt'). The assumption that the former is greater than the latter implies $\beta_i \leqslant \alpha_i$ $(0 \leqslant \beta_i < 1)$.

In contrast, the Bolton and Ockenfels (2000) inequality aversion model focuses on the relative position of the individual compared to the mean payoff of other players; note in the Fehr and Schmidt (1999) model, players focus on absolute differences. A simplified version of the Bolton–Ockenfels utility function has the following form (Wilkinson & Klaes, 2012):

$$U_i = U\left(x_i, \frac{x_i}{\sum x_j/n - 1}\right) \quad \text{where } j \neq i \quad (2)$$

Here, in addition to the absolute value of their earnings, players compare their payoff with the average payoff of others. In this model, players prefer a payoff that is nearer to the average.

The prediction from these two models of inequality aversion can be directly tested in a multivariate regression framework using our money burning experimental data. To test the Fehr and Schmidt (1999) model, we computed the difference between a player's payoff to that of each of the five members of his/her group within which money burning happens. Note that this variable is negative for players with lower payoff and positive for those with higher payoff – hence, the Fehr and Schmidt (1999) model predicts that money burning should fall with higher values of this variable, other things being equal. To test the Bolton and Ockenfels (2000) model, we computed the ratio of the payoff of a player to the average payoff of the other five players in the money burning group. Note that this ratio will be less (greater) than one if the player's payoff is lower (higher) than the average payoff of the other group members. Hence, the Bolton and Ockenfels (2000) model predicts money burning will fall with higher values of this ratio, other things being equal.[6]

The next section presents the main experimental results. Details on data collection are presented in the Appendix.

3. EXPERIMENTAL RESULTS

In this section we first provide a general overview of the results. We then test for different underlying motives for money burning and examine the relationship between money burning rates and real-life agricultural innovations.

(a) *General overview*

Money burning rates are defined as the amounts of money that subjects burn out of the total money available to burn (i.e., from each of the other players) in percentage terms. Table 1 shows the money burning rates for different groups and survey sites. Note that in each stage players make five pairs of money burning decisions (from safe money and lottery winnings)[7]; for 360 players and for three stages, this constitutes 5,400 pairs of money burning decisions. The figures provided in the first panel of Table 1 are related to these disaggregated money burning decisions. Using these disaggregated figures, the overall mean burning rate is around 8% and the corresponding median is zero; in 56% of the cases, players burned nothing even when there is some money to burn (in 280 cases there was no money to burn since players have used all their money on lottery and lost). That said, the maximum

Table 1. *Descriptive statistics*

	Mean	Median	Max	Std. dev.
Total money burning rates (%) (disaggregated money burning decisions)				
From total payoff	8.16	0	100.00	14.02
From safe money	7.93	0	100.00	15.45
From lottery winnings	9.02	0	100.00	15.78
Total money burning rates (%) (for individual players)				
From total payoff	8.24	6.56	64.18	8.85
From safe money	8.37	5.92	87.21	10.72
From lottery winnings	8.57	5.56	65.22	10.32
Total money burning rates (%)				
Stage 1	11.93	8.48	100.00	14.28
Stage 2	7.67	4.55	69.62	10.52
Stage 3	5.98	2.97	50.00	8.25
Total burning (Birr)				
Stage 1	7.31	5.00	57.00	8.70
Stage 2	4.93	3.00	59.00	7.17
Stage 3	4.09	2.00	35.00	5.72
Overall investment rate (%)	37.80	33.33	100.00	28.18
	Percentile	Centile	95% conf. interval	
Percentiles of total money burning rates (%) (for individual players)				
	10	0.00	0.00	0.00
	20	0.04	0.00	1.19
	30	2.58	1.44	3.44
	40	4.50	3.47	5.29
	50	6.56	5.26	7.48
	60	8.47	7.47	9.33
	70	10.26	9.39	11.40
	80	12.90	11.88	15.00
	90	18.68	16.73	22.72
	100	64.18	64.18	64.18
	Rural		Addis Ababa	
	Variable groups		Fixed groups	Variable groups
Mean money burning rates (%) (urban–rural and fixed and variable groups)				
From un-invested earnings	7.88		12.03	4.71
From lottery earnings	9.16		12.42	5.80
Overall	8.27		11.89	5.47
Mean investment rates on lottery (%)	28.68		49.13	62.93
Male (%)	65.00		93.33	88.33
Age (years)	46.24		21.51	21.35
No education (%)	49.58		0.00	0.00

burning rates of 100% (where players burn all available money) and the relatively high standard deviations indicate that there are significant numbers of high money burning rates.

The second and third sets of money burning rates in Table 1 are aggregated by individual players. Even though in the majority of separate money burning decisions, players did not burn anything (as we have seen above), the median figures are now significantly different from zero when we look at the average decisions of players; there are only 72 players from 360 who did not burn anything in all the three stages. This is also true if the player averages are disaggregated by stages. The percentile distribution of money burning rates given in the middle of Table 1 indicates the wide dispersion. Even though the average money burning rate is about 8%, around 20% of the players on the average burnt more than 18%. The histograms in Figure 1 show the distribution of money burning rates and absolute amounts of money burnt.

No significant differences in money burning rates between safe money and lottery winnings are observed. The money burning rates and the total money burnt decrease in subsequent stages of the game. The relatively low money burning coupled with the random dictator design may have played a role here; when people are not burnt in the first stage, they may tend to burn less in subsequent stages.

Average money burning rates in Addis Ababa are about 12% with fixed and about half as much with variable groups, with the rural rates in-between. There is more money burning within fixed than variable groups, as confirmed by a Mann–Whitney test on the Addis Ababa sample ($P = 0.05$). [8] On the other hand, there is no significant difference in money burning rates between variable groups in Addis Ababa and rural villages.

Result 1. Money burning rates are higher with fixed groups than with variable groups.

Players invest around 38% of their endowment on the lottery on average. This average figure hides the significant difference in investment rates between the university sample and the farmers in rural areas; while average investment rate of the former is 56% that of the latter is only 29%. The spread of investment

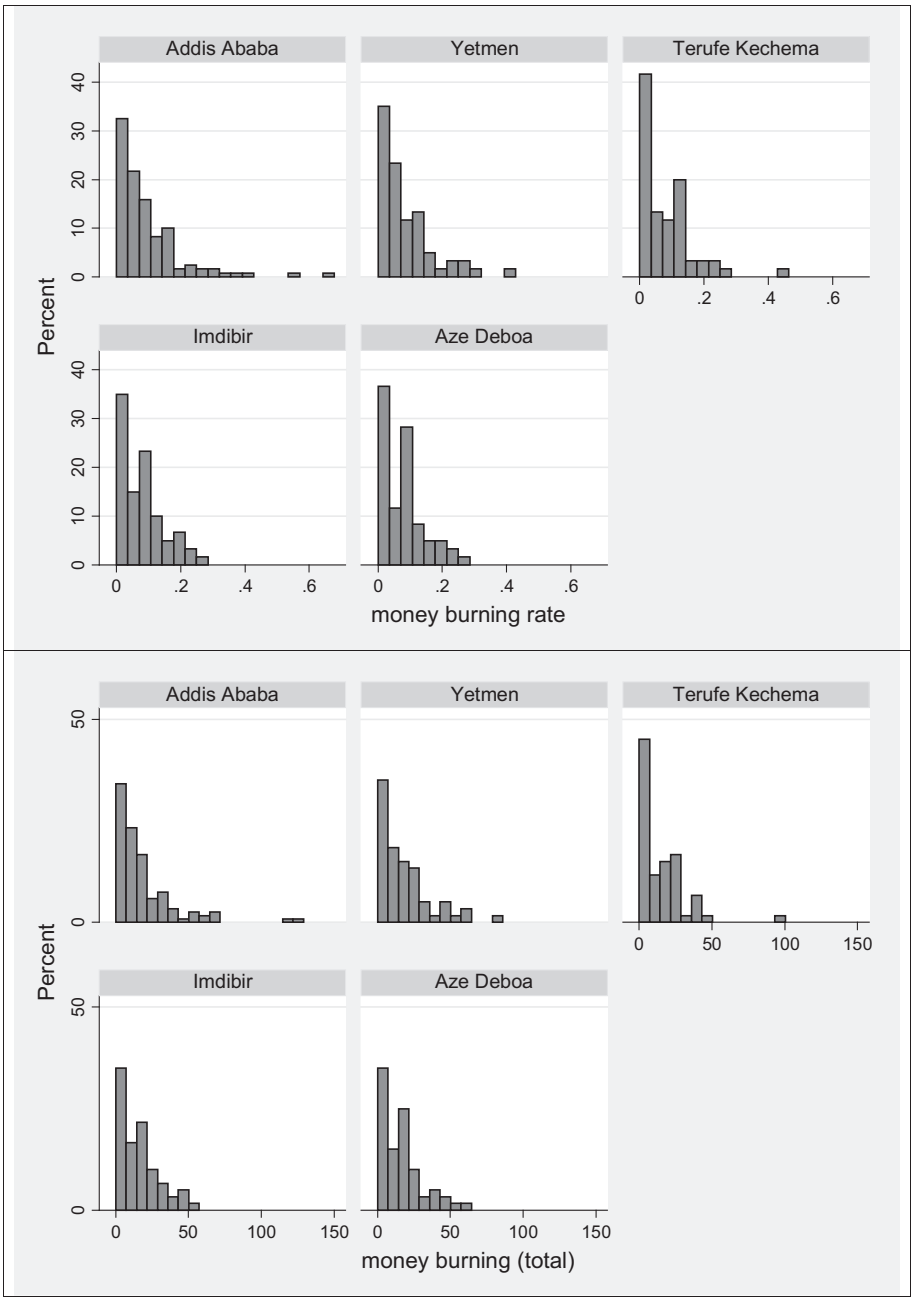


Figure 1. *Histograms of money burning rates (upper panel) and amount of money burning by sites (lower panel) (mean at player level for all three stages).*

rates and the difference between the urban and rural samples is clearly illustrated by the histograms in Figure 2.

The above figures imply that rural subjects are more risk averse than Addis Ababa students, as revealed by their choices to invest less in the lottery (Mann–Whitney $P < 0.001$). We suspect a combination of three factors likely explains this difference. First, there is evidence that risk aversion increases with age (e.g., Albert & Duffy, 2012); the average age of the rural sample is around 46 years with the corresponding figure for the students being 21 (look at Table 1). Second, there is evidence that risk aversion falls with increases in cognitive ability (e.g., Dohmen, Falk, Huffman, & Sunde, 2010). Particularly, in our case the comparison is between farmers who are largely uneducated (50% of whom are without any education) and students that are selected through highly competitive academic entrance procedures for their university education. Third, there is evidence that females are more risk averse than males (e.g., Borghans, Golsteyn, Heckman, & Meijers, 2009). In the rural sample, 65% are males; the proportion increases to 90% with university students. The dominance of males in the university sample is a third potential explanatory factor.

Result 2. Rural subjects are more risk averse.

The next section will shed more light on the possible underlying motives for money burning, particularly focusing on inequality aversion and reciprocity/retaliation.

(b) *Money burning, inequality aversion, and reciprocity*

To test the two inequality aversion models as described in Section 2(b), we used the disaggregated experimental game data. Note that, in each stage, each player is matched with five other players and is given information on what amount of

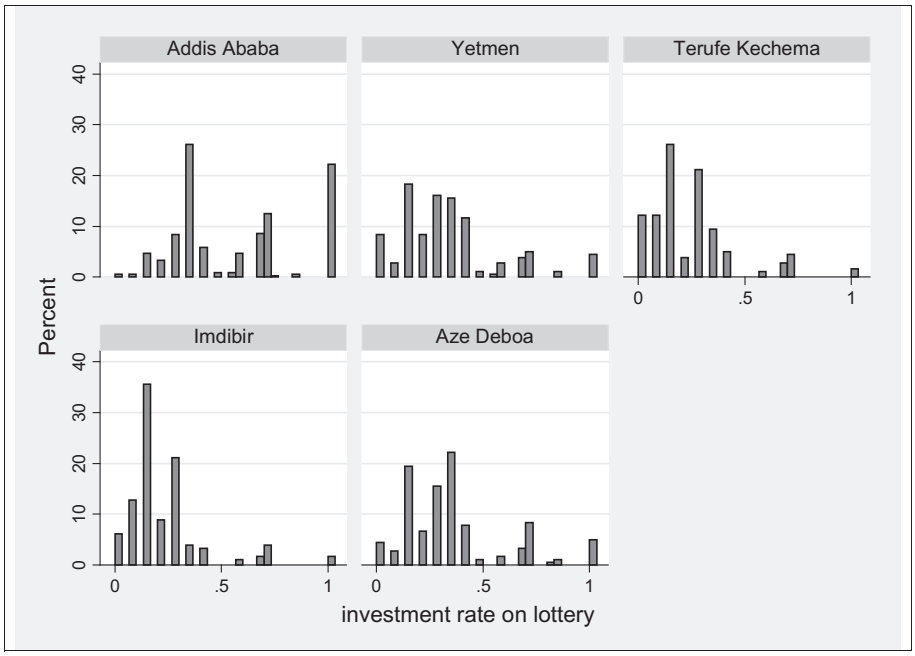


Figure 2. *Histograms of investment rates by sites (all rounds).*

money they have and is asked how much he/she would like to burn. In total there are 360 players and with three stages and five decisions of money burning for each stage the total number of observations adds up to 5,400 providing ample degrees of freedom. The money burning rate, as defined above, is the percentage of money a player would like to burn from the total money (safe money plus lottery winning) of another player in the group at that stage. The money burning rate is regressed on the above-defined inequality aversion variables, the total income of the money burner, whether the other player is a high- or low-income player and location and game related dummy variables. In a significant number of cases, players decided to burn nothing (money burning rate is zero); to handle the censoring problem we used panel tobit models with individual-level random effects. Four variations of the regressions are reported in Table 2. In the first and second models the absolute income difference and ratio of own income to average income of others are entered separately; in both, game-related variables and location dummies are included. In the third and fourth regressions, both inequality aversion variables are included; in the fourth, additional socio-demographic variables are controlled for.

The coefficients on absolute total income differences are consistently negative and highly significant; in other words, the higher the difference between the payoff of a player and other players, the higher the player burns their money. This supports the prediction of the Fehr and Schmidt (1999) inequality aversion model. The positive and consistently significant coefficient on high-income players also supports inequality aversion; players tend to burn more money from those that received higher amounts of initial money.

Result 3. Players burn more money when the absolute difference between their payoff and the payoff of others increases – this finding supports the predictions from Fehr and Schmidt (1999) inequality aversion model.

In contrast, all the coefficients on the ratio of own income to the average income of the others is consistently positive and highly significant. In other words, the data do not support the prediction from the Bolton and Ockenfels (2000) model; when the ratio of own payoff relative to that of other group members increases, players tend to burn more money, exactly the opposite of the prediction from the model. Part of the reason for this result is most likely related to an income effect. Note that money burning is a costly mechanism of redistributing income – players pay one-tenth of the value of the money they would like to burn. If players with a higher payoff burn more because they can 'afford' to do so (an income effect), it will make the coefficients positive since the ratio is positively correlated to the player's own payoff. This seems to be confirmed by the positive coefficient on the total income of the money burner in specification (1) which becomes no longer significant when the income ratio variable is added. This is as a consequence of multicollinearity between the two variables. Hence, in addition to the fact that income of the player directly enters into the ratio, the significance of the regression coefficients indicate the income ratio variable is partly capturing an income effect. [9] The income effect does not fully explain the positive coefficient.

Result 4. The data from the money burning game do not support the prediction of the Bolton and Ockenfels (2000) model of inequality aversion.

To summarize, while there is a robust support for the prediction of the Fehr and Schmidt (1999) model, the available evidence from the data does not support the Bolton and Ockenfels (2000) model. In fact, the coefficients suggest the opposite of what the Bolton and Ockenfels (2000) model predicts.

The other possible motive for money burning is reciprocity/retaliation. The money burning game used the random dictator design; from all the money burning decisions made by the six group members, the decision of only one is randomly selected and implemented. The actual amount of money burnt in the previous stage can trigger retaliatory money burning in the current stage. Particularly, this motive can be stronger in the fixed groups since the identity of the group members is kept the same in all stages; in other words, the person who has burnt your money will still be in your group in the next stage. But in the case of the variable groups, in every stage players are matched with other players and if there is a retaliatory motive it will of a 'generalized' nature; that is, people

Table 2. *Absolute and relative inequality aversion: Panel tobit models (individual level random effects)*

Variables	(1)	(2)	(3)	(4)
Absolute total income difference	−0.0036***		−0.0043***	−0.0042***
	(0.0005)		(0.0005)	(0.0005)
Ratio of own income to average of others		0.0342***	0.0519***	0.0509***
		(0.0084)	(0.0087)	(0.0089)
Total income of money burner	0.0053***	−0.0015	0.0014	0.0016
	(0.0007)	(0.0009)	(0.0010)	(0.0010)
High-income player (the other player)	0.0225***	0.0517***	0.0163**	0.0175**
	(0.0078)	(0.0066)	(0.0078)	(0.0079)
Socio-demographic variables				
Male (dummy)				−0.0031
				(0.0317)
Age (log)				0.0410
				(0.0546)
No education (dummy)				0.0088
				(0.0342)
Game related variables				
Afternoon session	0.0053	0.0094	0.0058	0.0078
	(0.0241)	(0.0241)	(0.0243)	(0.0253)
Stage 2	−0.0818***	−0.0784***	−0.0809***	−0.0835***
	(0.0078)	(0.0077)	(0.0077)	(0.0078)
Stages 3	−0.1270***	−0.1138***	−0.1219***	−0.1224***
	(0.0081)	(0.0080)	(0.0081)	(0.0082)
Variable groups	−0.1384***	−0.1267***	−0.1411***	−0.1385***
	(0.0423)	(0.0421)	(0.0426)	(0.0431)
Location dummies (Addis Ababa baseline)				
Yetmen	0.0900**	0.0882**	0.1040**	0.1005**
	(0.0423)	(0.0422)	(0.0428)	(0.0432)
Terufe Kechema	0.0728*	0.0594	0.0811*	0.0846*
	(0.0425)	(0.0423)	(0.0429)	(0.0436)
Imdibir	0.0969**	0.0843**	0.1064**	0.1403**
	(0.0423)	(0.0421)	(0.0427)	(0.0560)
Aze Deboa	0.0697	0.0660	0.0820*	0.1087*
	(0.0424)	(0.0422)	(0.0428)	(0.0563)
Constant	−0.0313	0.0002	−0.0407	−0.2011
	(0.0332)	(0.0327)	(0.0335)	(0.2041)
Sigma_u	0.2149***	0.2142***	0.2168***	0.2177***
	(0.0108)	(0.0107)	(0.0109)	(0.0110)
Sigma_e	0.1843***	0.1834***	0.1829***	0.1832***
	(0.0030)	(0.0030)	(0.0030)	(0.0030)
Observations	5,121	5,121	5,121	5,031
Number of players	360	360	360	354

Note: The dependent variable in these regressions is the money burning rate, i.e., the ratio of money burnt to total payoff (safe money plus lottery winnings). Standard errors in parentheses;
*** $p < 0.01$.
** $p < 0.05$.
* $p < 0.1$.

would start to burn more even though the person that has burnt their money may no more be in their group.

To test for the potential existence of a retaliatory motive, money burning rates are regressed on actual money burnt in the previous stage (lagged actual money burnt). If this motive is important in money burning, this variable will have a positive and significant coefficient. Note that since there is no lagged actual money burnt for the first stage, the regressions use data from only the second and third stages (look at Table 3).

In all the four specifications reported in Table 3, the coefficients on lagged actual money burnt are never significant. Particularly note that in the fourth specification we have added an interactive term between lagged actual money burnt and the dummy variable for variable groups. If retaliation was an important motive at least in the fixed groups, this interactive term would have become significant; the coefficient is not significant. Hence, retaliation does not seem to be an important motive for money burning also in the fixed group versions of the game.

Result 5. Reciprocity/retaliation is not an important motive behind money burning in our experiment. This is true even if we particularly focus on the fixed groups where the identity of the members is kept the same across stages of the game.

So far we looked at the underlying motive for money burning and found that the data support the prediction of the Fehr and Schmidt (1999) model which focuses on absolute income

Table 3. *Retaliatory motive for money burning: Panel tobit models (individual level random effects)*

Variables	(1)	(2)	(3)	(4)
Lagged actual money burnt	0.0032	0.0007	0.0009	−0.0014
	(0.0027)	(0.0028)	(0.0028)	(0.0053)
Lagged actual money burnt*variable group				0.0031
				(0.0061)
Total income of money burner	0.0001	0.0002	0.0003	0.0003
	(0.0007)	(0.0007)	(0.0007)	(0.0007)
High-income player (the other player)	0.0450***	0.0447***	0.0456***	0.0456***
	(0.0077)	(0.0077)	(0.0078)	(0.0078)
Male (dummy)			0.0049	0.0054
			(0.0325)	(0.0325)
Age (log)			0.0648	0.0641
			(0.0561)	(0.0561)
No education (dummy)			0.0021	0.0032
			(0.0351)	(0.0352)
Afternoon session		0.0142	0.0182	0.0316
		(0.0248)	(0.0258)	(0.0440)
Stages 2		0.0295***	0.0276***	0.0303
		(0.0083)	(0.0084)	(0.0443)
Variable groups		−0.0725*	−0.0675	0.0950*
		(0.0432)	(0.0438)	(0.0571)
Yetmen		0.0351	0.0318	0.0504
		(0.0435)	(0.0441)	(0.0577)
Terufe Kechema		0.0292	0.0309	0.0174
		(0.0436)	(0.0443)	(0.0258)
Imdibir		0.0507	0.0956*	0.0275***
		(0.0431)	(0.0572)	(0.0084)
Aze Deboa		0.0132	0.0502	−0.0715
		(0.0435)	(0.0577)	(0.0444)
Constant	−0.1172***	−0.0994***	−0.3557*	−0.3499*
	(0.0177)	(0.0346)	(0.2102)	(0.2103)
Sigma_u	0.2135***	0.2125***	0.2139***	0.2137***
	(0.0122)	(0.0122)	(0.0124)	(0.0124)
Sigma_e	0.1500***	0.1493***	0.1494***	0.1493***
	(0.0035)	(0.0035)	(0.0035)	(0.0035)
Observations	2,827	2,827	2,772	2,772
Number of players	360	360	354	354

Note: The dependent variable in these regressions is the money burning rate, i.e., the ratio of money burnt to total payoff (safe money plus lottery winnings). Standard errors in parentheses.
*$p < 0.1$.
**$p < 0.05$.
***$p < 0.01$.

differences. In contrast, the inequality aversion model that focuses on relative income comparison with the average income of other players as captured by the Bolton and Ockenfels (2000) model is not supported by the data. In addition, retaliation is not a significant motive for money burning.

In the next section, we examine if money burning captured in the game is related to real-life agricultural innovations.

(c) *Agricultural innovations and money burning*

One of the main objectives of this research is to understand how social preferences captured by money burning games are related to real-life agricultural innovations. We shall examine these issues first by combining the experimental data from the current project with household data from the Ethiopian Rural Household Survey (ERHS)[10] and second using additional data on agricultural innovations collected by the current research project. The use of two data sets generated by entirely independent data generating processes provides credence to our results. Note that the first part of this section uses data only from rural participants that were covered by the ERHS.

Money burning observed in the experimental game likely proxies for some unobserved social preference in real life and agricultural innovation is expected to be affected by these social preferences through three different channels.[11] First, individual money burning likely captures relevant individual characteristics correlated to innovation. For example, people burning others' money are likely more aggressive and/or competitive and hence we hypothesize that these characteristics may be correlated to innovative behavior. Second, the money burning of others (*social money burning*) is expected to affect innovation; a farmer in a community with high social money burning will be discouraged to invest as some of the returns from investment will be destroyed by others.[12] Third, the interaction between individual and social money burning is also an important factor. Even if a positive correlation

between individual money burning and innovation is expected, if everybody in the community behaves similarly the net effect may be different depending on feedback effects (externality).

Given other determinants, the innovation function can be presented as follows:

$$I_{hv} = f\left(MBR_{hv}{}^{h}, MBR^{v}_{hv}, MBR^{h}_{hv} \times MBR^{v}_{hv}/IN_{hv}, S_{sv}, V_{v}\right) \quad (3)$$

In Eqn. (3) the subscripts h and v index households and villages respectively; in our sample the households are from four villages hence v runs from 1 to 4. I_{hv} stands for innovations implemented by household h in village v, MBR^{h}_{hv} and MBR^{v}_{hv} stand for individual and social money burning rates and IN_{hv}, S_{sv} and V_{v} represent individual-, session-, and village-level effects that affect innovations. $MBR^{h}_{hv} \times MBR^{v}_{hv}$ stands for the interactive term between individual and social money burning rates included to capture the externality (feedback) effects.

To measure the overall adoption rates of households (I_{hv}) a simple innovation index is computed using the ERHS data. This index captures twelve agricultural innovations; farmers were asked whether they grow new crops like coffee and *chat*, [13]

Table 4. *Innovation index (from Ethiopian Rural Household Survey and from the data collected in this research project)*

(1) ERHS				(2) Money burning project			
Innovation index	Frequency	Percent	Cum. Freq.	Innovation index	Frequency	Percent	Cum. Freq.
1	19	12.67	12.67	0	42	21.65	21.65
2	21	14.00	26.67	1	57	29.38	51.03
3	30	20.00	46.67	2	76	39.18	90.21
4	29	19.33	66.00	3	19	9.79	100.00
5	21	14.00	80.00				
6	22	14.67	94.67				
7	6	4.00	98.67				
8	2	1.33	100.00				
Total	150	100.00		Total	194	100.00	

Table 5. *Innovations (ERHS) and money burning: Multi-level regressions (site and session levels random effects)*

Variables	(1)	(2)	(3)	(4)
Investment rates on lottery game	0.0522	−0.0602	0.0809	−0.1220
	(0.6176)	(0.6173)	(0.6133)	(0.6356)
Money burning rate from safe money	−0.7805	−4.5530**	−41.0097**	−31.7879
	(1.7274)	(1.9655)	(20.4330)	(20.2959)
Money burning rate from lottery win	0.3017	−4.0992***	13.9574	15.3235
	(1.4215)	(1.5724)	(16.1412)	(15.5910)
Safe money social burning rate for PA		−224.4653***	−257.6613***	−250.0944***
		(51.5005)	(55.5914)	(60.4536)
Lottery win social burning rate for PA		−258.0384***	−241.1391***	−222.4717***
		(43.7726)	(46.9992)	(51.1965)
Individual*social money burning rates (safe money)			478.5592*	370.9191
			(264.1420)	(261.1661)
Individual* social money burning rates (lottery win)			−199.7049	−219.0262
			(181.0975)	(174.8396)
Male (dummy)				0.7638***
				(0.2395)
Age (log)				−0.2122
				(0.3942)
No education (dummy)				0.0326
				(0.2363)
Constant	3.9321***	45.4322***	46.4347***	44.4326***
	(0.6375)	(7.2030)	(7.6368)	(8.3208)
Study site var(_cons)	0.1791	−1.1291**	−1.1183**	−0.9836**
	(0.3639)	(0.4995)	(0.4925)	(0.4589)
Session var(_cons)	−25.7324***	−14.3079	−14.3235	−17.1825**
	(8.0622)	(2,368.6676)	(2,284.4102)	(7.3763)
Var(Residual)	0.2301***	0.2301***	0.2137***	0.1769***
	(0.0585)	(0.0585)	(0.0585)	(0.0587)
Observations	150	150	150	149
Number of groups	4	4	4	4

Note: The dependent variable in these regressions is the innovation index computed from the Ethiopian Rural Household Survey (ERHS) in relation to each subject which took part in the ERHS. Standard errors in parentheses.
*** $p < 0.01$,
** $p < 0.05$,
* $p < 0.1$.

whether they have improved livestock, use modern agricultural inputs, farm other new crops, use irrigation, water holes, improved seeds and/or fertilizer, whether they were selected as model farmers, participated in soil conservation, or rain harvesting programs. If their responses are affirmative a score of 1 is given for each of these innovations; if not, they get a score of 0. The simple sum of these scores provides an index of innovation. For example, a farm household that adopted all the twelve innovations will have a score of twelve and a household that has not adopted any will have a score of zero. [14]

Households on average have adopted around four of the innovations (mean = 3.75) and at least one innovation has been adopted by all households with the most innovative households implementing eight (look at the figures under (1) ERHS in Table 4).

Community level *social money burning rates* (MBR^{v}_{hv}) are computed in the following way. For each individual the average money burning rates of all *other* individuals (i.e., excluding the individual's money burning rate) in the village is computed. Social effects should reflect what each individual expects on the average from the rest of the people in the village.

To estimate Eqn. (1) the innovation index is regressed on individual and social money burning rates and the interaction between the two in addition to other controls. The regressions are estimated using multi-level mixed effects models with survey site and session level random effects to control for unobservable effects at those levels. The same model is estimated by including different variables to see if results are robust to inclusion and exclusion of variables.

In most cases, individual money burning rates – both from safe money and lottery wins – are not significant. In the three cases they become significant, they have the opposite sign to our initial expectation. There is no robust correlation between the individual money burning rates and their investment in real agricultural innovations.

The highly significant, consistent, and robust result from the regressions reported in Table 5 is related to social money burning rates for both safe money and lottery winnings. For both variables and for all specifications, the coefficients on the social money burning rates are highly significant and as expected negative. The results strongly suggest that money burning in the experimental games likely captures an important aspect of social preference which is inimical to agricultural innovations particularly on the community level. In communities where the money burning rate is high, famers innovate significantly less. On the other hand, the interactive terms between individual and social money burning are not significant (only one being significant merely at 10% level). This result suggests there is no strong feedback (externality) effect in money burning behavior.

Note that the detailed information on agricultural innovations used so far comes from the ERHS, a survey that was

Table 6. *Innovations (fertilizer, improved seeds, and rain harvesting) and money burning: Multi-level regressions (site and session levels random effects)*

	(1)	(2)	(3)
Investment rates on lottery game	0.6957**	0.6593*	0.4121
	(0.3408)	(0.3424)	(0.3439)
Money burning rate from safe money	−2.1188**	9.5834	12.7837
	(1.0207)	(11.7617)	(11.2962)
Money burning rate from lottery win	−0.4684	6.7307	7.5390
	(0.8341)	(8.9468)	(8.4291)
Safe money social burning rate for PA	−61.2423***	−48.7662**	−49.9196**
	(20.2190)	(23.1956)	(23.5538)
Lottery win social burning rate for PA	−89.8015***	−81.9200***	−72.7667***
	(17.3827)	(19.6132)	(19.9101)
Individual*social money burning rates (safe money)		−150.6836	−185.9199
		(151.8993)	(145.4998)
Individual* social money burning rates (lottery win)		−82.5129	−94.4218
		(101.1747)	(95.2679)
Male (dummy)			0.5175***
			(0.1252)
Age (log)			0.1764
			(0.2084)
No education (dummy)			−0.2093*
			(0.1261)
Constant	14.2640***	12.5871***	10.9913***
	(2.8276)	(3.1176)	(3.1988)
		−2.4458**	−2.2344***
Study site var(_cons)	−2.3564**	(1.1753)	(0.7286)
	(1.0002)	−2.9982	−24.3847***
Session var(_cons)	−3.3579	(4.2436)	(9.1970)
	(8.3410)	−0.2182***	−0.2798***
Var(Residual)	−0.2145***	(0.0520)	(0.0514)
	(0.0520)		
Observations	194	194	193
Number of groups	4	4	4

Note: The dependent variable in these regressions is the innovation index computed from the data set collected by the current research project. Standard errors in parentheses.
*** $p < 0.01$.
** $p < 0.05$.
* $p < 0.1$.

conducted independently of the current research. In the current research project, data on agricultural innovations, particularly focusing on the use of fertilizer, improved seeds, and rain harvesting, were collected. Similar to the above procedure, we computed a simple innovation index capturing the adoption of the three innovations by households; for each innovation adopted by the households, a score of 1 is given otherwise 0. Hence, households adopting all three will have an innovation index of 3 and those who have not adopted any will have a score of 0. The mean innovation index for the sample is 1.37, meaning that on the average households have adopted less than two innovations. Around a fifth of the sampled households have not adopted any one of the three (figures under '(2) Money burning project' in Table 4).

As in the previous case, this new innovation index is regressed on individual and social money burning rates and their interaction in addition to some control variables. Multi-level mixed models with site and session levels random effects are estimated (look at Table 6).

Like in the previous case, individual money burning rates are not significantly correlated to the innovation index. All the social money burning rates for both safe money and lottery winnings on the other hand are highly significant and negative. The results from the previous regressions are confirmed. As indicated before, since the ERHS and the current project generated these data sets independently, one can be more confident on this robust result. Social money burning on the community level is robustly and negatively correlated to agricultural innovation as illustrated from the results of the two independent data sets. As in the previous case, the interaction terms between individual and social money burning rates are not significant.

Result 6. Higher social money burning rates at the village level are negatively and robustly correlated to agricultural innovations measured by indices based on the ERHS as well as data collected by the current research. This implies that the money burning behavior captured in the experimental games captures social preferences that are inimical to real-life agricultural innovations.

So far, indices that capture the overall number of innovations from two independent data sets are used as measures of agricultural innovations. An alternative is to look at individual innovations. In Table 7, regression results from multi-level probit models using specific innovations (fertilizer, improved seeds, and rain harvesting) are presented. The results from these probits generally support the results from innovation indices. In all specifications, individual money burning from safe money as well as from lottery winnings is not significantly correlated to the specific innovations. Similar to the previous results, most of the coefficients on the social money burning rates from safe money and lottery wins are negative and significant. From the twelve coefficients seven are highly significant and negative; in all the twelve cases coefficients are consistently negative. It is interesting to note that

Table 7. *Specific innovations and money burning: Multi-level probit regressions*

Variables	(1) Fertilizer	(2) Improved seeds	(3) Rain harvesting	(4) Fertilizer	(5) Improved seeds	(6) Rain harvesting
Investment rates on lottery game	0.7168	1.8672***	−0.4564	0.3924	1.7913**	−0.8087
	(0.7423)	(0.6326)	(0.7904)	(0.8826)	(0.7104)	(0.8702)
Money burning rate from safe money	1.7007	15.4791	20.1726	6.9161	29.5874	25.3777
	(38.5041)	(20.8804)	(24.1502)	(43.1485)	(23.3745)	(25.1445)
Money burning rate from lottery win	19.3796	15.4288	−3.3495	20.8923	20.7466	−4.3190
	(27.2593)	(16.5533)	(22.3195)	(29.6692)	(18.1861)	(22.8312)
Safe money social burning rate for PA	−6.0111	−78.9846**	−43.0317	−9.9455	−87.8358*	−42.2550
	(59.3255)	(40.0564)	(39.9752)	(71.7952)	(47.3210)	(41.0427)
Lottery win social burning rate for PA	−128.4483***	−67.9062**	−126.7073**	−132.2366***	−55.0200	−135.4135***
	(41.0269)	(34.4172)	(49.5069)	(50.9432)	(40.1073)	(52.3205)
Individual*social m. burning rates (safe money)	−26.8987	−244.9709	−301.8285	−89.7684	−414.7643	−367.0172
	(509.6125)	(267.5987)	(314.7038)	(570.6215)	(298.3167)	(326.8493)
Individual* social m. burning rates (lottery win)	−232.5595	−173.0377	34.7478	−254.4901	−239.3083	40.0589
	(295.6409)	(186.2598)	(261.3861)	(320.1827)	(204.2425)	(267.3850)
Male (dummy)				0.7473**	1.1305***	0.0424
				(0.3313)	(0.2488)	(0.3053)
Age (log)				0.7745	0.3450	−0.4131
				(0.4853)	(0.3811)	(0.4934)
No education (dummy)				−0.5729*	−0.2820	−0.1163
				(0.3036)	(0.2335)	(0.2948)
Constant	12.9974*	11.9547**	13.7894**	10.7096	9.5303	16.2208**
	(6.6510)	(5.4718)	(6.7375)	(8.3843)	(6.4720)	(7.2627)
var(_cons[site])	0.0441	0.0000	0.0000	0.0907	0.0549	0.0000
	(0.0702)	(0.0000)	(0.0000)	(0.1270)	(0.0715)	(0.0000)
var(_cons[site>session])	0.0000	0.0584	0.0000	0.0121	0.0000	0.0000
	(0.0000)	(0.0764)	(0.0000)	(0.1116)	(0.0000)	(0.0000)
Observations	194	194	194	193	193	193
Number of groups	4	4	4	4	4	4

Note: The dependent variable in these regressions is a dummy variable representing adoption of the given innovation based on the data set collected by the current research project. Standard errors in parentheses.
*** $p < 0.01$.
** $p < 0.05$.
* $p < 0.1$.

all but one of the coefficients on social money burning from lottery wins are significant. The weight of the evidence in this section clearly shows a strong and robust negative correlation between social money burning rates and agricultural innovations. Some of the evidence for this is derived using data that were collected by another survey independently of the current research; hence, this provides strong support to the results. Contrary to our expectation, individual money burning rates are not significantly correlated to innovation activities. In addition, the interaction between individual and social money burning behavior does not create an external feedback effect.

As indicated before, sociological reports were prepared by university lecturers and eight focus group discussions were conducted as part of this research in addition to the experimental game. The qualitative data from these sources strongly suggest behavior similar to money burning in real-life activities. For example, envy is considered by most as an automatic consequence of success and achievement, as illustrated by the farmer's quote in the introduction. The sociological reports provide stories of destructive behavior even among close family members. [15]

In this paper we focused on agricultural innovations; but note that farmers are also involved in non-agricultural innovations. In the long-term, since dependence on agriculture is expected to decline with economic growth, non-agricultural innovations are expected to become more important. Some of the anecdotal evidence from the qualitative data indicates that sabotaging behavior becomes even stronger in relation to activities that are very different from farming. Extension agents indicate that the problem is stronger when innovations are very novel (Tiumelissan, 2009). Since this research focused only on agricultural innovations, the problem is likely to be more severe if non-agricultural innovations are considered. At a general level, growth of non-agricultural sectors is crucial in the structural transformation of economies and sabotaging (and fear of sabotaging) can become an important obstacle to development.

4. CONCLUSION

We presented an experiment on money burning primarily run with Ethiopian farmers. Money burning rates were lower than observed without the social pressures of the field environment. While the data support the inequality aversion model of Fehr and Schmidt (1999), there is no evidence for the alternative inequality aversion model of Bolton and Ockenfels (2000). Reciprocal/retaliatory motives do not play an important role in our experiment. Furthermore, money burning, particularly money burning on the level of communities, is robustly and negatively correlated to agricultural innovations. To our knowledge, this is the first paper that captures money burning behavior using experimental games and examines their effect on real-life innovations.

There are three take home messages for policy makers from this research. First, sabotaging behavior proxied in this research by money burning matters for adoption, even when controlling for a number of other variables. Second, while changing preferences may be difficult, there may be institutional changes that can be made to help channel such preferences in a productive rather than a destructive direction, as argued by Grolleau, Mzoughi, and Sutan (2009). Third, the negative effect may be minimized if innovations are adopted by significantly large number of people in the community at early stages. The usual model of small number of adopters followed by the majority later may not be effective; a 'big push' of innovation may be required to break a sort of low equilibrium trap created by the fear of sabotaging activities. Obviously further research is needed.

NOTES

1. In the same spirit, the book by Bowles and Herbert (2011) is titled 'A cooperative species' and focuses on pro-social behavior.

2. We are grateful to one of the reviewers for suggesting these references.

3. Birr is the national currency of Ethiopia. At the time of the games, US$1 was worth around Birr 8.

4. This random dictator design follows Zizzo (2003). Note that, if the money burning decisions of all the participants are implemented, potentially the money to be burnt could be more than the money given to players.

5. The scripts of instructions for the variable and fixed groups are provided in an Online Appendix.

6. For the Bolton and Ockenfels (2000) model it does not matter how the money burning is spread across the other players *as long as* the ratio is improved toward the average payoff of the other players. This is why, by having both a subject-specific payoff difference and a ratio variable based on the average payoff of the other players as independent variables in a regression analysis, we will be able to respectively identify a Fehr and Schmidt (1999) or a Bolton and Ockenfels (2000) model effect.

7. The logic for making a distinction between safe and lottery winnings is based on the fact that safe money is kept without any risk but lottery money is earned after going for a risky undertaking.

8. Here and below in this section, we consider our tests at the level of independent observations: this means the session in the case of variable groups, and the group in the case of fixed groups. For example, in this Mann–Whitney test, $n = 10$ for the fixed groups and $n = 2$ for the variable group sessions. The regression analysis will employ the data more efficiently while controlling for the potential non independence of observations. P values in this paper are two-sided throughout.

9. To assess whether the income of the money burner would be significant in the absence of multicollinearity, we did the following. The variance inflation factor (VIF) for total income of the money burner in the regression reported in column (4) is 4.45; the square root of the VIF (2.11) indicates by how much the standard error is inflated due to multicollinearity. If the standard error of the coefficient is divided by the square root of the VIF, we get 0.0005 which is an estimation of what would be the standard error if there was no multicollinearity. With this standard error the income of the money burner would have become significant. This is additional evidence reinforcing our claim that the income ratio is partly capturing an income effect.

10. The ERHS is a panel household survey which has been conducted in 15 rural villages starting from 1994 in all important agro-ecological regions of the country. This paper uses the data from the round conducted in 2004; previous rounds were run in 1994, 1995, 1997, and 1999.

11. Here we are focusing on social preferences measured from our experimental games but are not discounting the very wide literature on agricultural innovations.

12. It is possible that innovation may simply be discouraged by the underlying inequality aversion but it is also possible that the burning specifically of money obtained after going for a risky undertaking may be a better proxy for the discouragement of real-world innovations. Because of this we distinguish between money burning rate out of safe money and money burning out of lottery winnings.

13. *Chat* (*Catha edulis*) is a mildly intoxicating plant that is consumed widely both in Ethiopia as well as neighboring countries. It has become an important source of income for farmers as well as a major foreign exchange earner for the country.

14. Obviously, this innovation index is a very rough measure. Among its other limitations, it does not take into account substitutability and complementarity between the different innovations.

15. In Imdibir, one of the study villages, a man is reported to have set fire to his brother's farm when his brother started cultivating a more profitable cereal.

16. *Enset* (*Ensete ventricosum*) is the false banana tree and is used as staple food among many ethnic groups in southern Ethiopia.

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APPENDIX A. DATA COLLECTION AND SURVEY SITES

The key data for this research are gathered from four villages. The villages were covered in a previous panel household survey which provides detailed data going as far back as 1994. One of the villages, Imdibir, is located southwest of the capital Addis Ababa in the Gurage region, a region in the heartland of a farming system based on *enset* [16] that is characterized

by the use of the hoe rather than the ox-plough. The majority of the people are members of the Gurage, a highly entrepreneurial ethnic group active in commerce. The second, village Aze Deboa, is in the Kembata region, dominated by an ethnic group of the same name. Even though *enset* is important, cereals and other crops play a more prominent role compared to Imdibir. The third village, Terufe Kechema is near one of the biggest commercial centers in the south, the town of Shashemene. The region is dominated by the Oromo, the most populous ethnic group in Ethiopia. The fourth village, Yetmen, is located northwest of Addis Ababa and like most regions in the north farming is dominated by the cultivation of cereals and use of the ox plough. Yetmen and the surrounding area are well known for the production of *teff*, a type of cereal used for human consumption only in Ethiopia. The strong demand for *teff* has made the region relatively prosperous. As the short description above indicates, important ethnic and farming system differences are captured by these villages. Two of the most populous ethnic groups in Ethiopia – Oromo and Amhara – are the dominant ethnic groups in the two sites, Terufe Kechema and Yetmen, respectively. Similarly, the villages also capture important variations in agricultural systems: the ox-plough *versus* hoe-culture, *enset versus* cereals. The spread of agricultural innovations is also varied; for example, the percentages of farmers using fertilizer range from a low of 45% in Imdibir to a high of 97% in Yetmen.

Even though the main data came from the money burning experimental game, additional information was also gathered. Pre-game questionnaires capture background information on participants, agricultural practices, and innovations. Post-game questionnaires gathered information on the players' experience of the game. While the above-mentioned research instruments were administered on individuals participating in the game, additional information was also gathered through focus group discussions and sociological surveys. In each village, four focus groups – two with males and two with females – were organized. Four sociology lecturers from Addis Ababa University were commissioned to write sociological reports on each village using both gray literature and field visits where key informant interviews, local materials, folklores, anecdotal evidences, and similar information were gathered.

The rural fieldwork was conducted in February–March 2009. Due to low mobility/migration in rural Ethiopia, tracking households that were interviewed in the last round of the rural household survey in 2004 was not too difficult. While 60 individuals were required for the two sessions of the game in each village, the household survey covered more than that in each village. Some individuals (33 out of 240, i.e., 14%) did not turn up at the time of the game and had to be replaced by others not originally covered by the household survey.

The games were also played with 120 Addis Ababa University students in February 2010. The students were mainly at the undergraduate level and from the Education and Business and Economics faculties. While two sessions at Addis Ababa University are identical to the rural ones, two sessions with fixed groups of players were also conducted; this was done to examine the effect of playing with the same individuals in the three stages on money burning behavior as well as increase the number of independent observations and improve statistical power.

In three villages, halls that accommodate players sitting reasonably far apart were used; in the remaining one village the game was played in two big tents. In Addis Ababa, a very big hall that can accommodate a large number of people was used. Research assistants were intensively trained and one of the authors of this paper oversaw the conduct of all the experimental sessions.

In all cases, participants came to the designated place so that we had a more controlled environment compared to playing the games sequentially at different times and places like the homes of players. But significant differences with experimental game laboratories in Western universities should be noted. In the rural areas all individuals are drawn from the same village. Even though players could not identify who their group members are, each participant sees the remaining 29 players most or all of whom are likely to be known to him/her. Similarly, university students were drawn from a small number of faculties and hence are likely to know each other. This is different from complete anonymity in proper experimental labs where individuals interact through computer terminals. The lack of complete anonymity likely increases pro-social behavior, and so we believe the money burning rates are likely to be biased downward.

APPENDIX B. SUPPLEMENTARY DATA

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.worlddev.2014.10.022.

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